

REVISION RECORD			
VER	REV NO.	APPROVED:	DATE:

TABLE LIST

Sheet #	Content	Sheet #	Content
01.	Table of Content	31.	SKY77916+SKY77643
02.	Revision History	32.	LTE_DUPLEXER
03.	Block Diagram	33.	LTE_B7
04.	GPIO MAP	34.	DRX
05.	PMI8952_CHARGER	35.	QFE2101
06.	PMI8952_CONTROL/INTERFACE	36.	WCN3615
07.	PM8937_CONTROL_AND_MPP/CLOCK		
08.	PM8937_BUCK_CONVERTER		
09.	PM8937_LDO_CIRCUITS		
10.	PM8937_CODEC		
11.	MSM8937_CONTROL		
12.	MSM8937:_EBI		
13.	MSM8937:_GPIO		
14.	MSM8937:_MIPI_AND_RF_CONTROL		
15.	MSM8937:_POWER1		
16.	MSM8937:_POWER2		
17.	MSM8937:GND		
18.	MEMORY:LPDDR3+EMMC		
19.	BATTERY_CONNECTOR		
20.	CHARGE		
21.	MIC_RECEIVER&SPEAKER		
22.	Headset		
23.	LCD_INTERFACE_AND_BACKLIGHT		
24.	MAIN/SLAVE_CAMERA_AND_FLASH		
25.	Sensors		
26.	SIM/TF_CARD		
27.	KEYPAD/RGB_LED_DRIVER		
28.	NC		
29.	TEST_POINT/GND/SHIELDS		
30.	WTR2965		

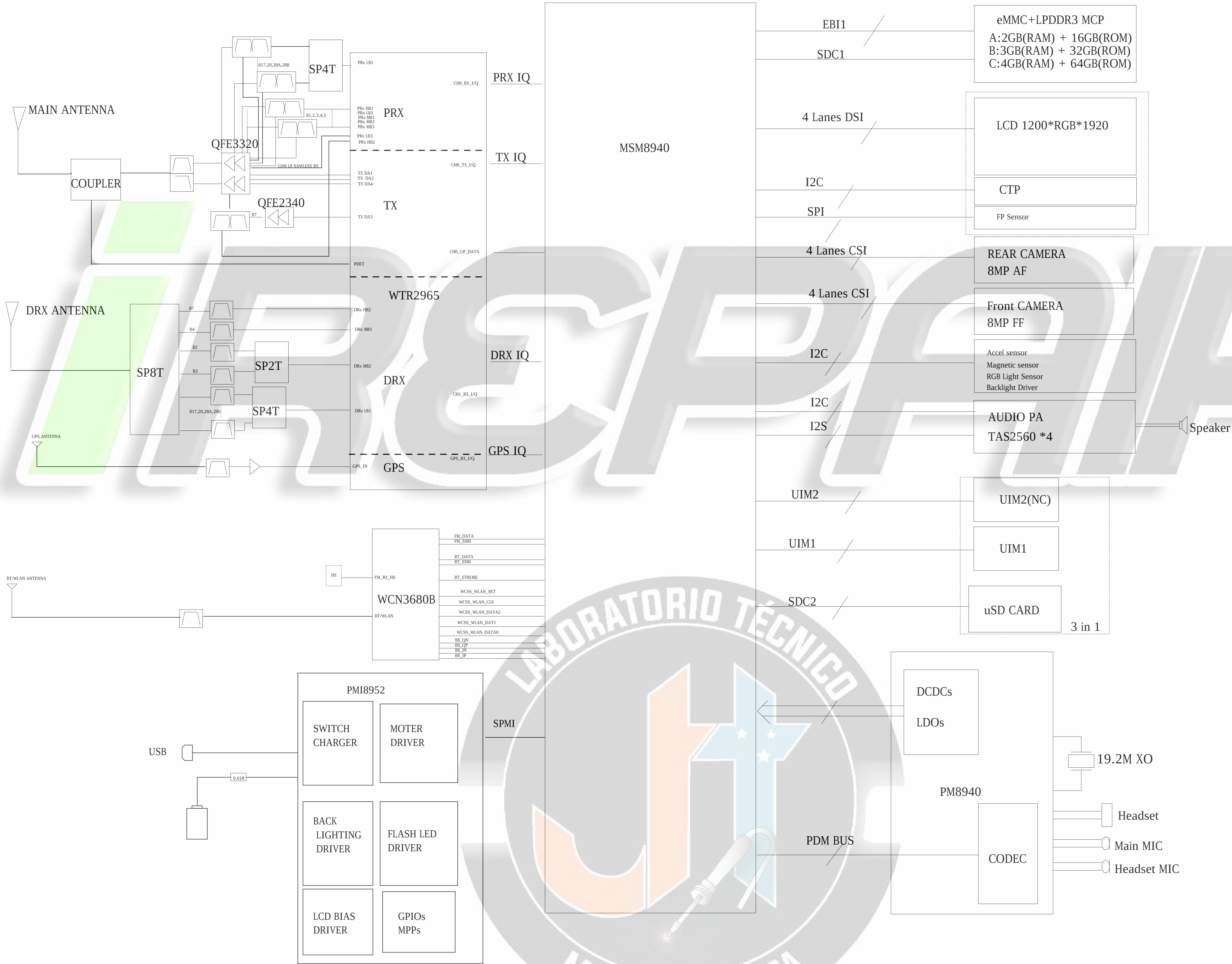


COMPANY: Shanghai Longcheer Technology Co., Ltd			
TITLE: TABLE LIST			
CODE:	SIZE:	DRAWING NO:	REV:
XX	A0	XXXX	A

DRAWN:	DATED:
zhangzhiqiang	20160926
CHECKED:	DATED:
yaogang	20160926

REVISION RECORD			
LTB	ECO NO.	APPROVED	DATE

BLOCK

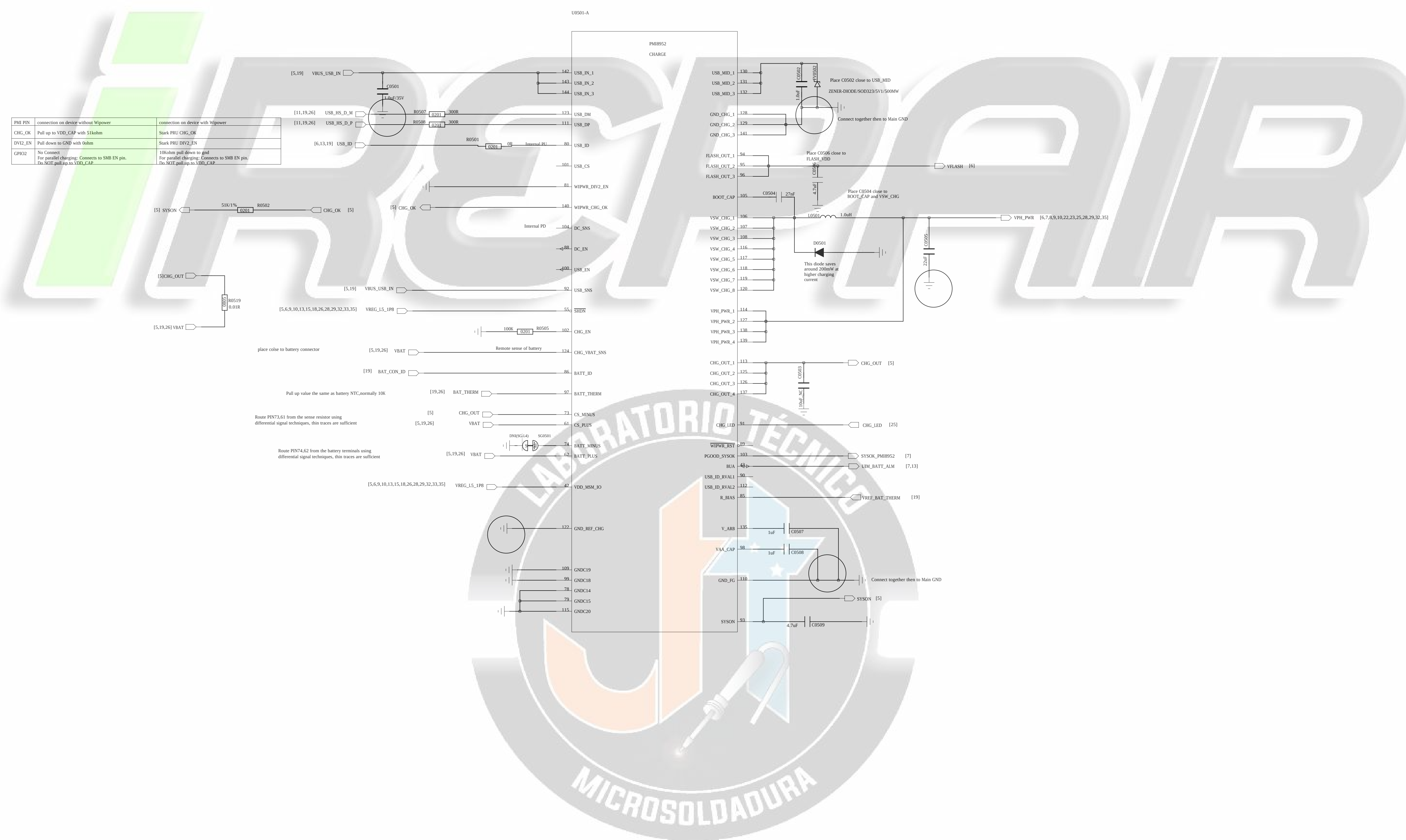


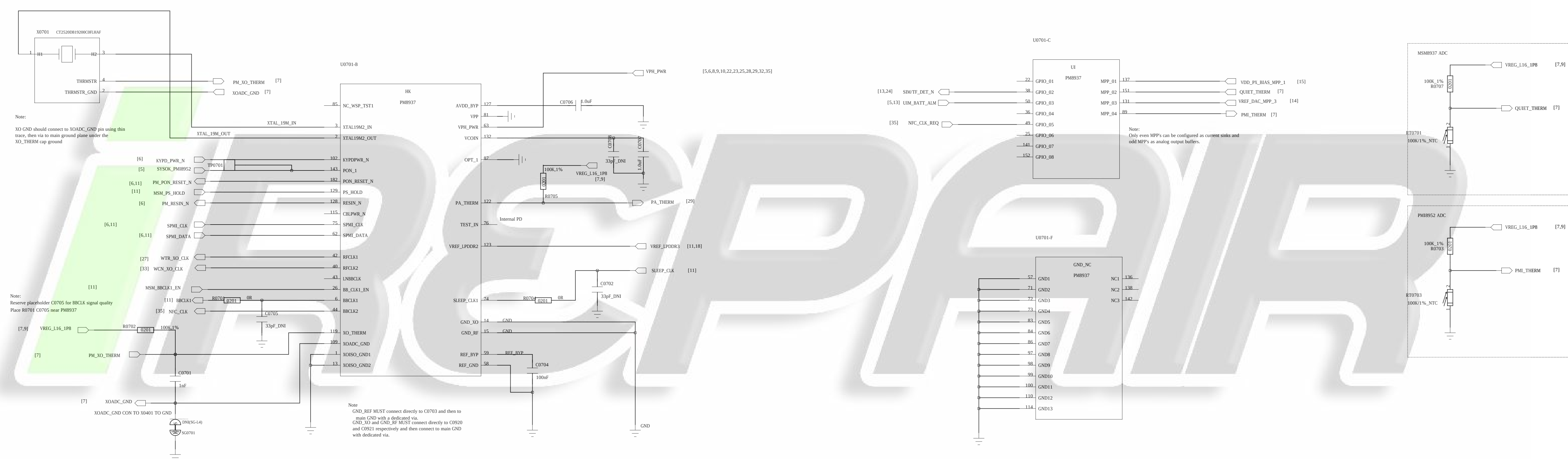
GPIO MAP

REVISION RECORD			
VER	ECO NO.	APPROVED	DATE

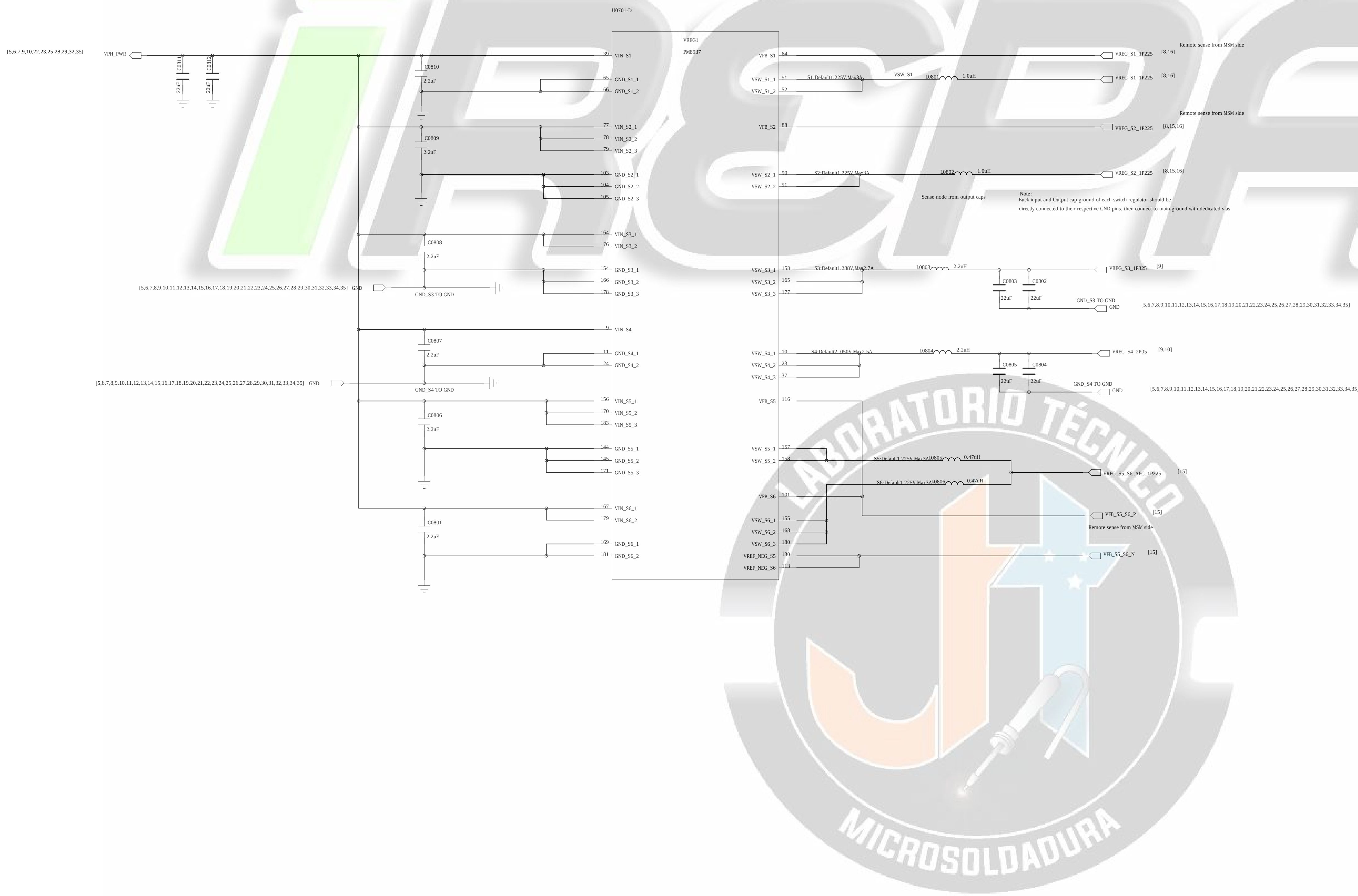
MSM8940 GPIO Configuration For QRD8909							
GPIO_0	NC	GPIO_41	LIGHT_INT1	GPIO_82	NC	GPIO_123	NC
GPIO_1	NC	GPIO_42	ACCL_INT1_N	GPIO_83	WCSS_BT_CTRL	GPIO_124	AUDIO_INT4
GPIO_2	BL/LIGHT_SDA	GPIO_43	LIGHT_INT2	GPIO_84	WCSS_BT_DAT_STB	GPIO_125	GPIO_DOME_SWITCH
GPIO_3	BL/LIGHT_SCL	GPIO_44	MAG_RST_N	GPIO_85	MI2S_1_SCK	GPIO_126	NC
GPIO_4	UART_MSM_TX	GPIO_45	ACCL_INT2_N	GPIO_86	MI2S_1_D1	GPIO_127	HALL_EINT1
GPIO_5	UART_MSM_RX	GPIO_46	FP_INT_N	GPIO_87	MI2S_1_WS	GPIO_128	AUDIO_INT2
GPIO_6	SPK_SDA	GPIO_47	CAM_2V8_EN	GPIO_88	MI2S_1_D0	GPIO_129	GPIO_FP_ID
GPIO_7	SPK_SCL	GPIO_48	GPIO_BOX_ID	GPIO_89	BOARD_ID0	GPIO_130	AUDIO_INT2
GPIO_8	NC	GPIO_49	UIM_BATT_ALM	GPIO_90	BOARD_ID1	GPIO_131	GPIO_BOX_ID2
GPIO_9	NC	GPIO_50	FP_RST_N	GPIO_91	KEY_VOLP_UP_N	GPIO_132	GPIO_ICM_LED_EN
GPIO_10	TP_I2C_SDA	GPIO_51	UIM1_DATA	GPIO_92	NC	GPIO_133	NC
GPIO_11	TP_I2C_SCL	GPIO_52	UIM1_CLK	GPIO_93	AUDIO_INT3		
GPIO_12	MI2S_2_D0	GPIO_53	UIM1_RESET	GPIO_94	MI2S_2_SCK		
GPIO_13	MI2S_2_D1	GPIO_54	SIM/TF_DET_N	GPIO_95	MI2S_2_WS		
GPIO_14	BUCK_SMB_I2C_SDA	GPIO_55	NC	GPIO_96	BOARD_ID2		
GPIO_15	SENSOR_I2C_SCL	GPIO_56	AUDIO_RST	GPIO_97	BOARD_ID3		
GPIO_16	CAPSENSOR_RDY	GPIO_57	NC	GPIO_98	NC		
GPIO_17	CTP_1V8_EN	GPIO_58	NC	GPIO_99	NC		
GPIO_18	CAPSENSOR_SDA	GPIO_59	GPIO_ICM_LED_EN	GPIO_100	RFFE1_CLK		
GPIO_19	CAPSENSOR_SCL	GPIO_60	NC	GPIO_101	RFFE1_DATA		
GPIO_20	FP_SPI_MOSI	GPIO_61	NC	GPIO_102	RFFE2_CLK		
GPIO_21	FP_SPI_MISO	GPIO_62	AUDIO_INT1	GPIO_103	RFFE2_DATA		
GPIO_22	FP_SPI_CS	GPIO_63	NC	GPIO_104	TX_CPL		
GPIO_23	FP_SPI_CLK	GPIO_64	TP_RST_N	GPIO_105	PRX_B2B34B39_SEL2		
GPIO_24	NC	GPIO_65	TP_INT_N	GPIO_106	LNA_B3		
GPIO_25	GPIO_CHG_DISABLE	GPIO_66	GP_PDM_A0	GPIO_107	PRX_ANT_TUNER_SW2		
GPIO_26	CAM0_MCLK	GPIO_67	SIM/TF_DET_N	GPIO_108	PRX_ANT_TUNER_SW1		
GPIO_27	NC	GPIO_68	FP_RST_N	GPIO_109	LNA_B7		
GPIO_28	CAM_MCLK1	GPIO_69	CDC_PDM_CLK	GPIO_110	PRX_B17B2B_SW1		
GPIO_29	CAM_I2C_SDA	GPIO_70	CDC_PDM_SYNC	GPIO_111	PRX_B17B2B_SW2		
GPIO_30	CAM_I2C_SCL	GPIO_71	CDC_PDM_TX	GPIO_112	PRX_SW_B40B41		
GPIO_31	CAM_I2C_SDA1	GPIO_72	CDC_PDM_RX0	GPIO_113	DRX_BAND		
GPIO_32	CAM_I2C_SCL1	GPIO_73	CDC_PDM_RX1	GPIO_114	DRX_B34B39_SW		
GPIO_33	LCM_3V3_EN	GPIO_74	CDC_PDM_RX2	GPIO_115	ANT_TUNING_EN		
GPIO_34	TP_3V3_EN	GPIO_75	WCSS_BT_SSB1	GPIO_116	DRX_B5B8_SW		
GPIO_35	MCAM_PWDN	GPIO_76	WCSS_WLAN_DATA_2	GPIO_117	PRX_SW_B1B4		
GPIO_36	MCAM_RST_N	GPIO_77	WCSS_WLAN_DATA_1	GPIO_118	EXT_GPS_LNA_EN		
GPIO_37	FORCE_USB_BOOT	GPIO_78	WCSS_WLAN_DATA_0	GPIO_119	CH0_GSM_TX_PHASE_D0		
GPIO_38	GPIO_ICM_LOADSW	GPIO_79	WCSS_WLAN_SET	GPIO_120	RFFE5_CLK		
GPIO_39	SCAM_PWDN	GPIO_80	WCSS_WLAN_CLK	GPIO_121	RFFE5_DATA		
GPIO_40	SCAM_RST_N	GPIO_81	NC	GPIO_122	NC		

PM8940 GPIO/MPP Configuration For MSM8940			
GPIO_1	NC	MPP_1	VDD_PX_BIAS_MPP_1
GPIO_2	SIM/TF_DET_N	MPP_2	USB_CONN_ADC
GPIO_3	UIM_BATT_ALM	MPP_3	VREF_DAC_MPP_3
GPIO_4	NC	MPP_4	QUIET_THERM





REVISION RECORD			
VER	ECO NO.	APPROVED	DATE



COMPANY: Shanghai Longcheer Technology Co., Ltd			
TITLE: PM8940-BUCK			
CODE: XX	SIZE: A0	DRAWING NO: XXXX	REV: A

DRAWN: zhangzhiqiang	DATED: 20160926
CHECKED: yaogang	DATED: 20160926

SCALE: <Scale>	SHEET: 8 OF 36
----------------	----------------

REVISION RECORD			
LTR	ECO NO.	APPROVED	DATE

PM8940-LDO

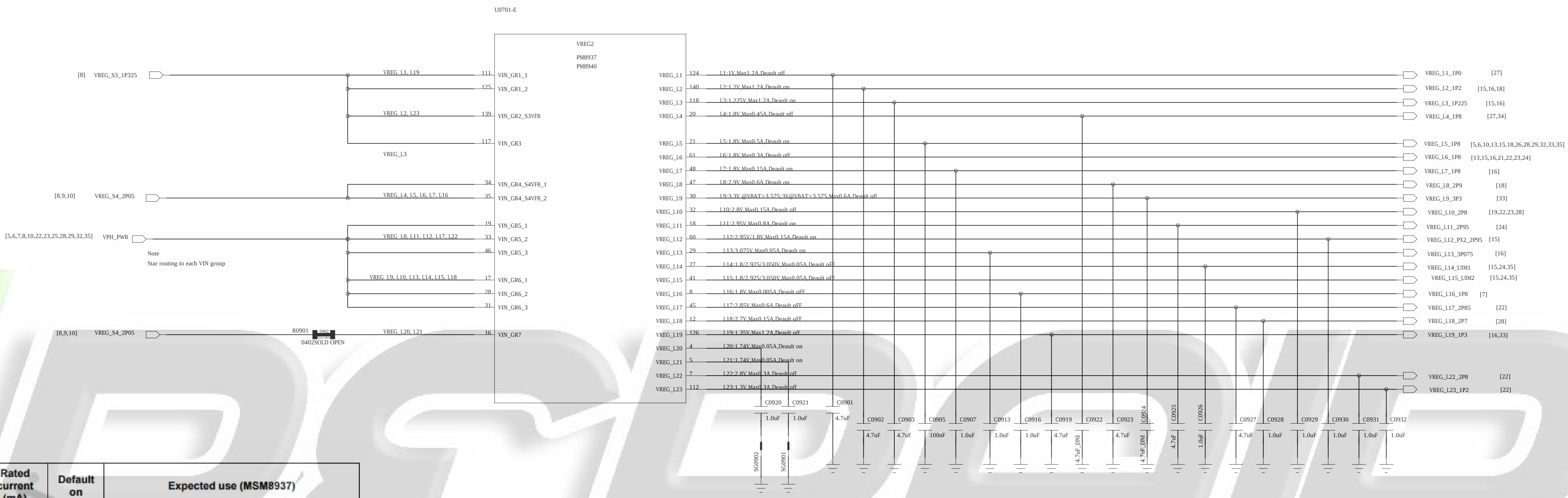


Table 3-6 PM8937/PM8940 regulators and their intended uses

Function	Circuit type	Default V (V) ¹	Specified range (V) ² (MSM8937)	Programmable range (V)	Rated current (mA)	Default on	Expected use (MSM8937)
S1	ULT-SMPS	1.225	0.900~1.350	0.375~1.5625	2000 ³	N	MSM modem
S2	ULT-SMPS	1.225	0.550~1.350	0.375~1.5625	3000	Y	MSM core and graphics
S3	HF-SMPS	1.288	1.200~1.4125	0.375~1.5625	2700	Y	Low-voltage LDOs (1, 2, 3, 19, and 23)
S4	ULT-SMPS	2.050	1.800~2.050	1.550~3.125	2500	Y	High-voltage LDOs (4, 5, 6, 7, 16, RFCLK, and XO)
S5	FT-SMPS	1.225	1.050~1.350	0.350~1.355	3000	Y	MSM applications processor
S6	FT-SMPS	1.225	1.050~1.350	0.350~1.355	3000	N	MSM applications processor
L1	NMOS LDO	1.000	1.000	0.375~1.5375	1200	N	RFICs
L2	NMOS LDO	1.200	1.200	0.375~1.5375	1200	Y	LPDDR2/LPDDR3, MIPI CSI, and DSI
L3	NMOS LDO	1.225	0.750~1.360	0.375~1.5375	1200	Y	VDDMX
L4	PMOS LDO	1.800	1.800	1.750~3.3375	450	N	RFICs and GPS eLNA
L5 ⁴	PMOS LDO	1.800	1.800	1.750~3.3375	500	Y	Most digital I/Os, MSM pad groups 3 and 7, LPDDR, and eMMC
L6	PMOS LDO	1.800	1.800	1.750~3.3375	300	N	MSM DSI PLL and OTP, camera, touchscreen, display, and sensors
L7	PMOS LDO	1.800	1.800	1.750~3.3375	150	Y	MSM analog and PLLs, WCN XO, and PM baseband clock driver
L8	PMOS LDO	2.900	2.900	1.750~3.3375	600	Y	eMMC
L9	PMOS LDO	$V_{out} = 3.3\text{ V}$ for $V_{BAT} > 3.575\text{ V}$, $V_{out} = 3\text{ V}$ for $V_{BAT} < 3.575\text{ V}$	3.000~3.300	1.750~3.3375	600	N	WCN
L10	PMOS LDO	2.8	2.800	1.750~3.3375	150	N	Sensors
L11 ⁵	PMOS LDO	2.950	2.950	1.750~3.3375	800	Y	Micro SD
L12 ⁴	PMOS LDO	2.950	1.800/2.950	1.750~3.3375	150	Y	MSM pad group 2 and SDC2
L13	PMOS LDO	3.075	3.075	1.750~3.3375	50	Y	MSM USB and audio

Function	Circuit type	Default V (V) ¹	Specified range (V) ² (MSM8937)	Programmable range (V)	Rated current (mA)	Default on	Expected use (MSM8937)
L14 ⁵	PMOS LDO	1.800	1.800/2.925/3.050	1.750~3.3375	50	N	MSM pad group 5, dual-voltage UIM1, and NFC
L15 ⁵	PMOS LDO	1.800	1.800/2.925/3.050	1.750~3.3375	50	N	MSM pad group 6 and dual-voltage UIM2
L16	PMOS LDO	1.800	1.800	1.750~3.3375	5	N	PMIC HKADC
L17	PMOS LDO	2.850	2.850	1.750~3.3375	600	N	Camera, display, and touchscreen
L18	PMOS LDO	2.700	2.700	1.750~3.3375	150	N	QTI RF front-end
L19	NMOS LDO	1.350	1.350	0.375~1.5375	1200	N	MSM analog, WCN, and WGR
L20	Low-noise LDO	1.74	1.74	1.74~3.3375	5	Y	PMIC XO circuits
L21	Low-noise LDO	1.74	1.74	1.74~3.3375	5	Y	PMIC RF clock buffers
L22	PMOS LDO	2.800	2.800	1.750~3.3375	300	N	Camera – analog
L23	NMOS LDO	1.3	1.2	0.375~1.5375	300	N	Camera – digital

PSEUDO CAPLESS LDOs

L4*, L6, L8*, L9*, L10, L11*, L12, L14, L15, L17*, L18, L22, L23

L4*, L8*, L9*, L11*, L17*, still need validation as Pseudo-capless

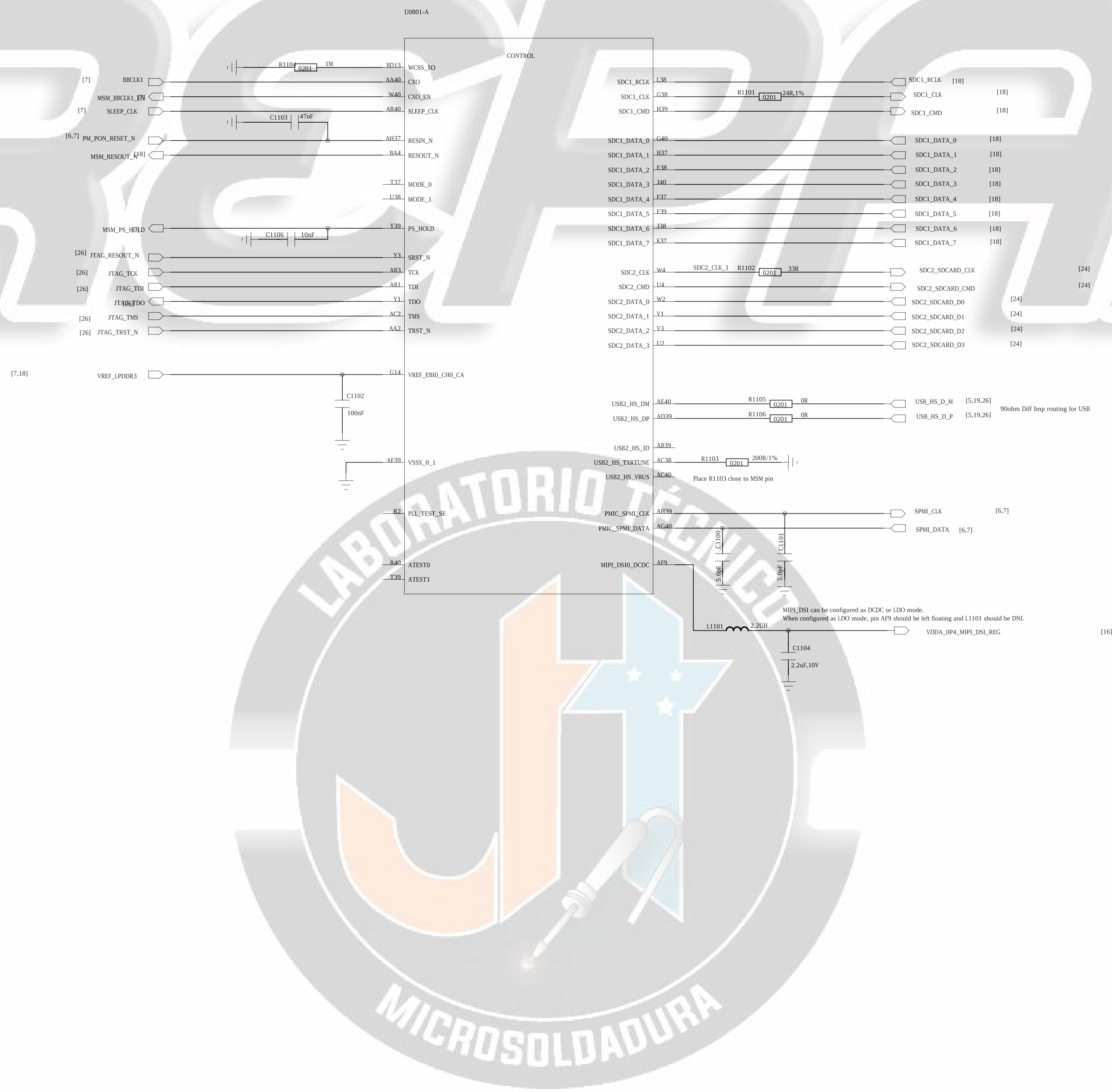
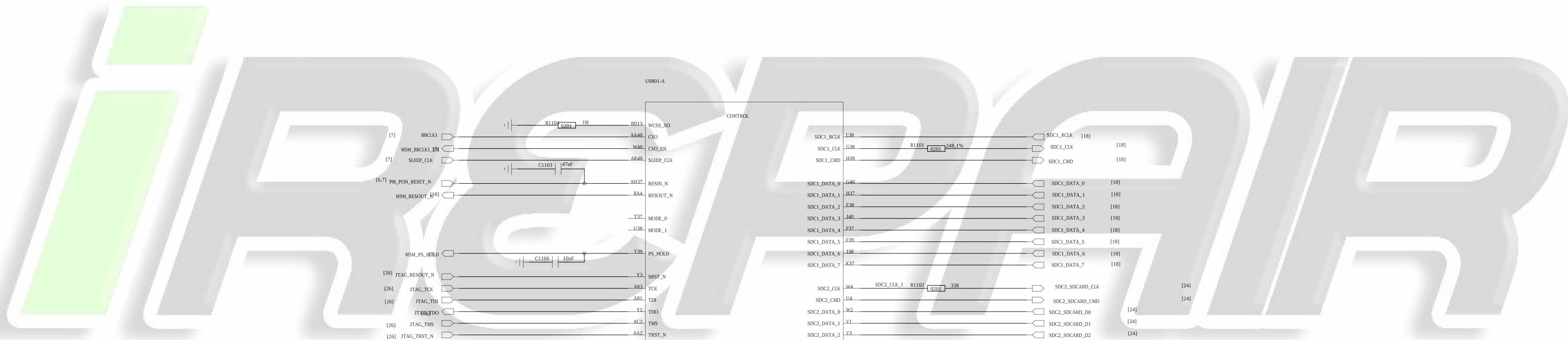
L5, L7, L13, L16 have internal PMIC Loads and require local CAPs.



COMPANY: Shanghai Longcheer Technology Co., Ltd			
TITLE: PM8940-LDO			
CODE: XX	SIZE: A0	DRAWING NO: XXXX	REV: A
SCALE: <Scale>			
SHEET OF 9 OF 36			

DRAWN: zhangzhiqiang	DATE: 20160926
CHECKED: yaogang	DATE: 20160926

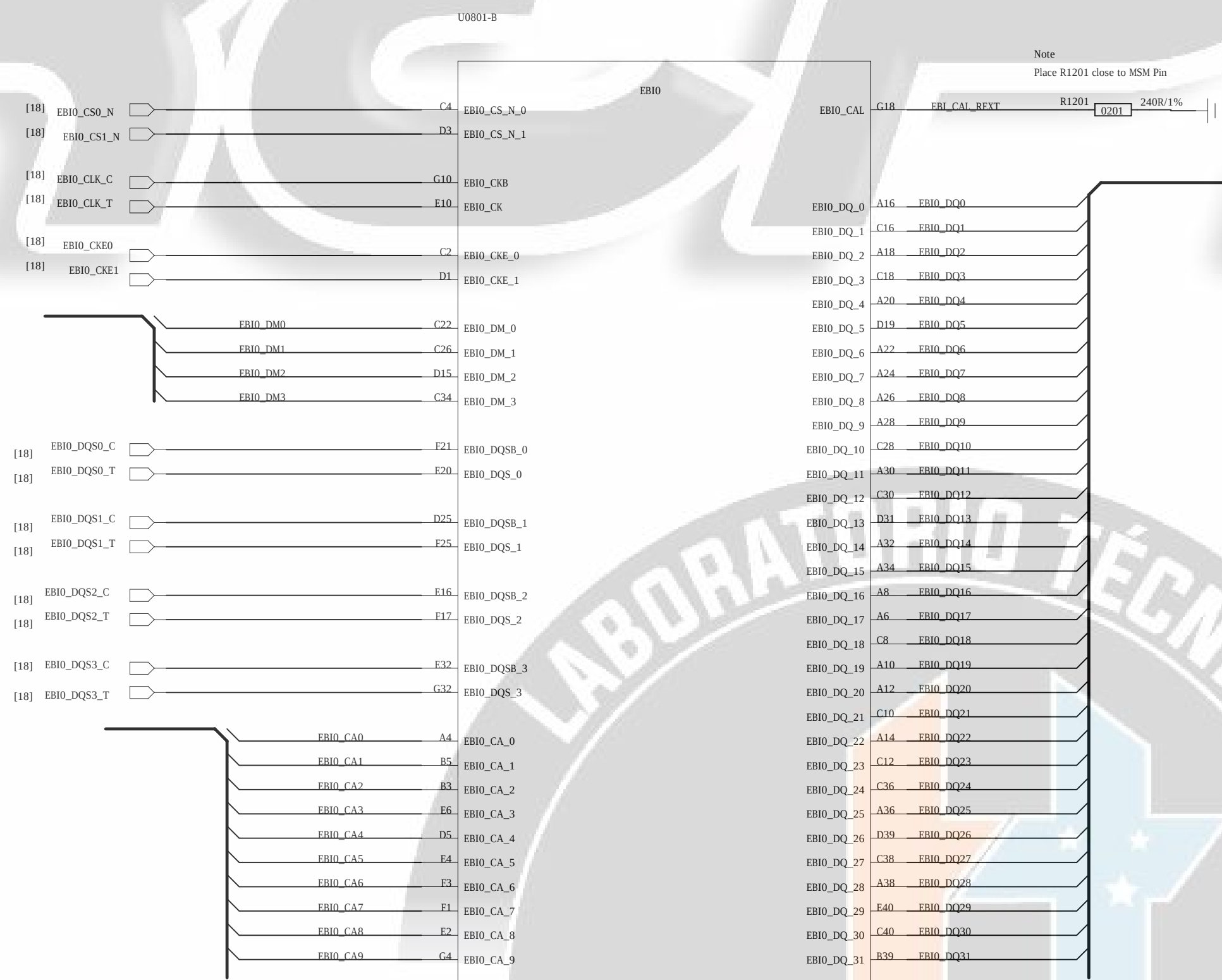
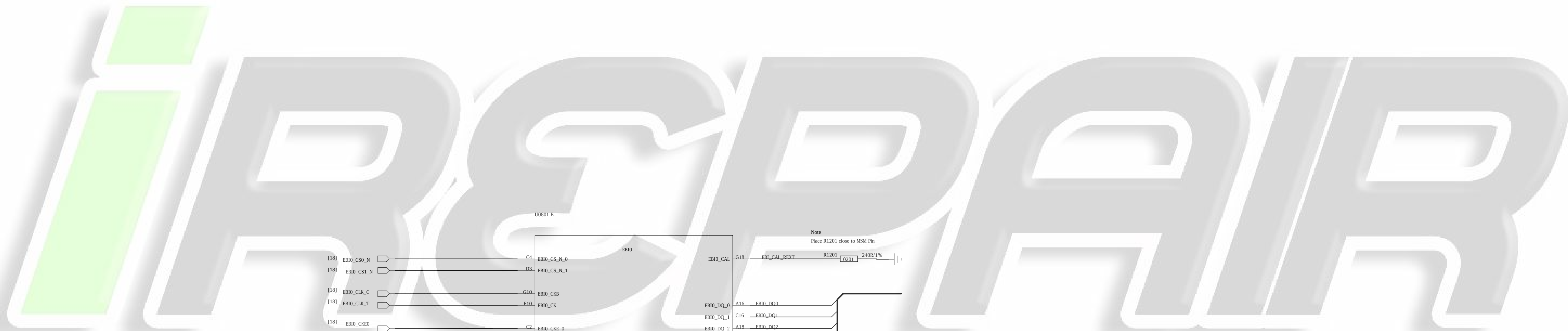
REVISION RECORD			
VER	REV NO.	APPROVED:	DATE:



COMPANY: Shanghai Longcheer Technology Co., Ltd			
TITLE: MSM8940-CONTROL			
CODE: XX	SIZE: A0	DRAWING NO: XXXX	REV: A
SCALE: <Scale>			
SHEET: OF 11 OF 36			

DRAWN: zhangzhiqiang	DATE: 20160926
CHECKED: yaogang	DATE: 20160926

REVISION RECORD			
LTB	ECO NO.	APPROVED	DATE



COMPANY: Shanghai Longcheer Technology Co., Ltd			
TITLE: MSM8940-EMI			
CODE:	SIZE:	DRAWING NO:	REV:
XX	A0	XXXX	A

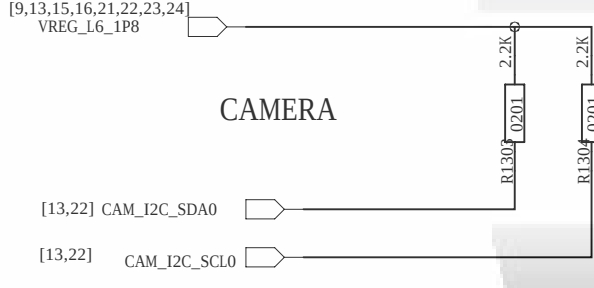
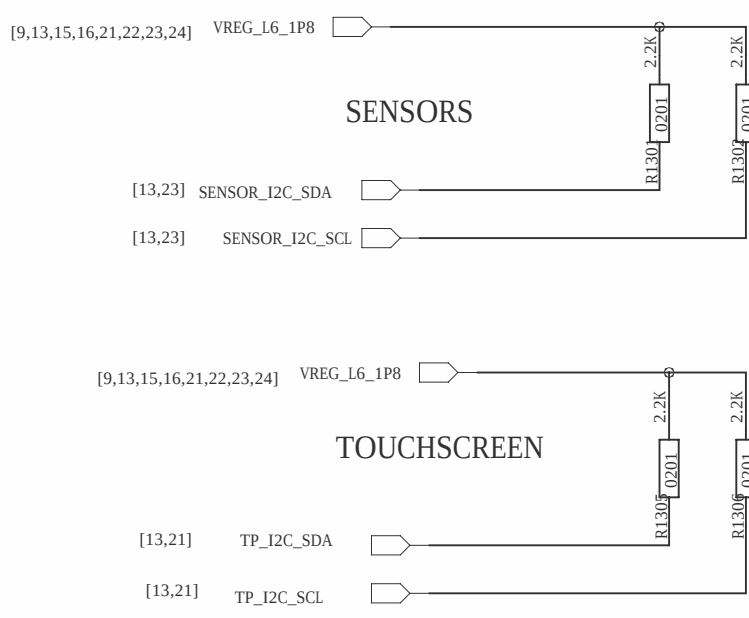
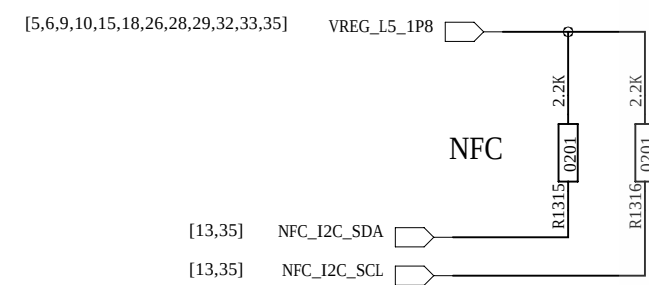
DESIGNER:	DATE:
zhangzhiqiang	20160926
CHECKER:	DATE:
yaogang	20160926

REVISION RECORD			
VER	ECO NO.	APPROVED	DATE



Note: Do not have pull-ups on the following GPIOs unless intend for boot or secure- boot related configurations:
GPIO_91, 107, 109

Note: Ensure SW sets these GPIOs (Sensor, CTP and Camera I2C bus) to input pull down when the peripherals are powered off to eliminate leakage.

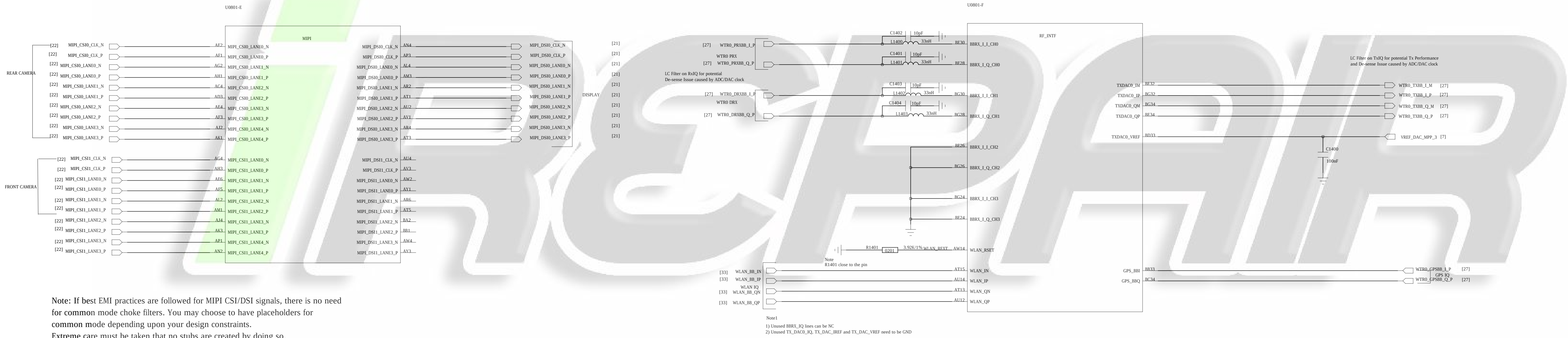


GPIO	BOOT_CONFIG
GPIO_3	BOOT_CONFIG[0]MODE_DISABLE
GPIO_111	BOOT_CONFIG[1]
GPIO_112	BOOT_CONFIG[2]
GPIO_98	BOOT_CONFIG[3]

BOOT_CONFIG[3:1]	BOOT_CONFIG
0x000	SDC1 -> SDC2 -> U9801-D
0x010	SDC1
0x0100	SDC2 -> SDC1

Default Boot Config (0x0000) is SDC1(0x010)

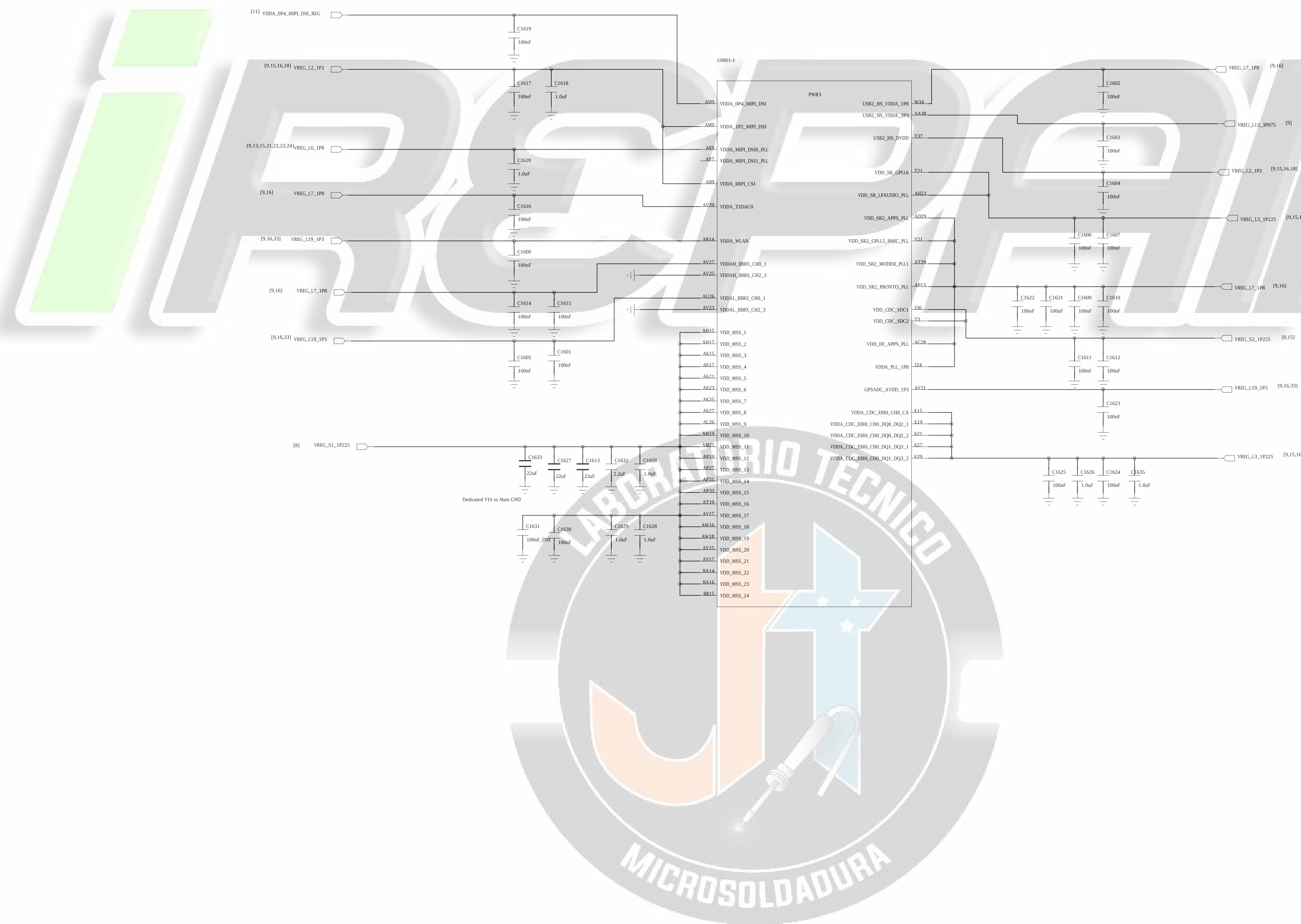
REVISION RECORD			
ITEM	ECO NO.	APPROVED:	DATE:



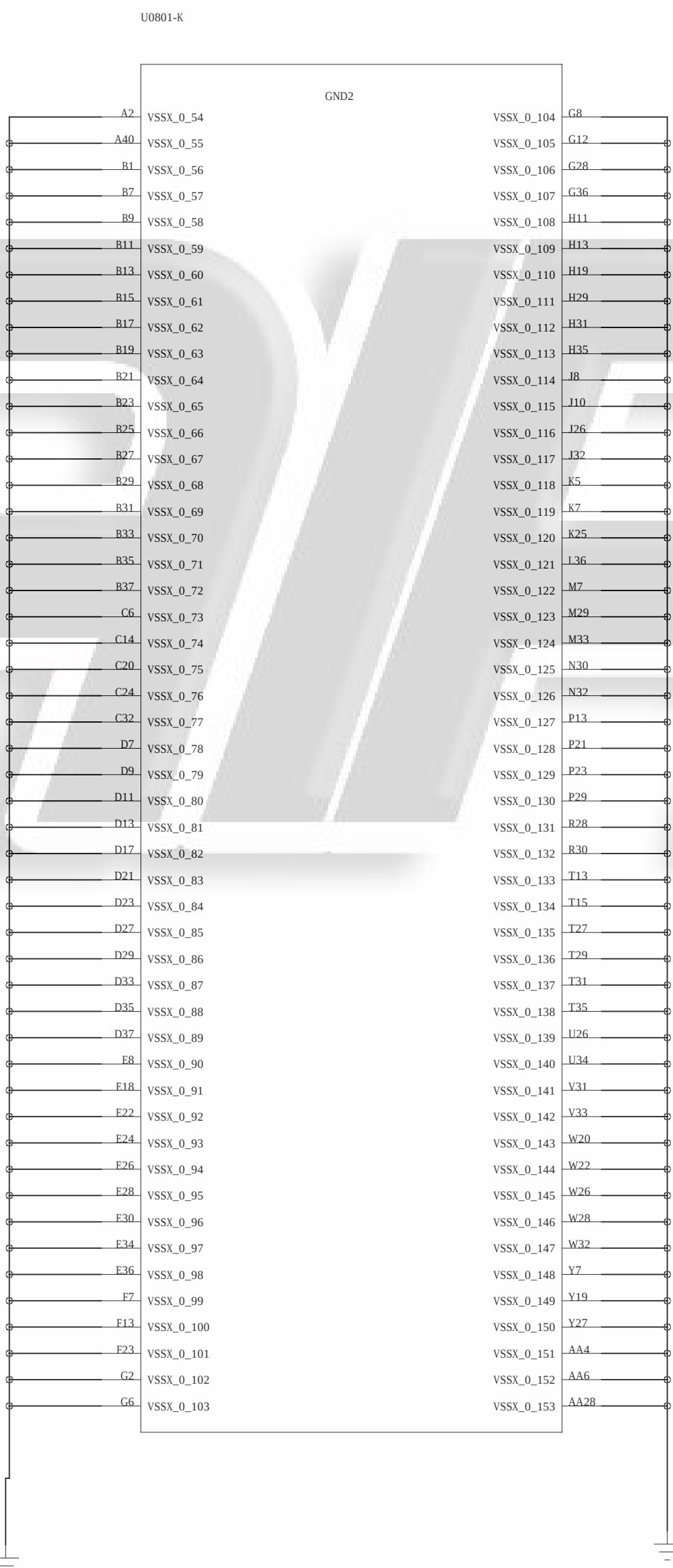
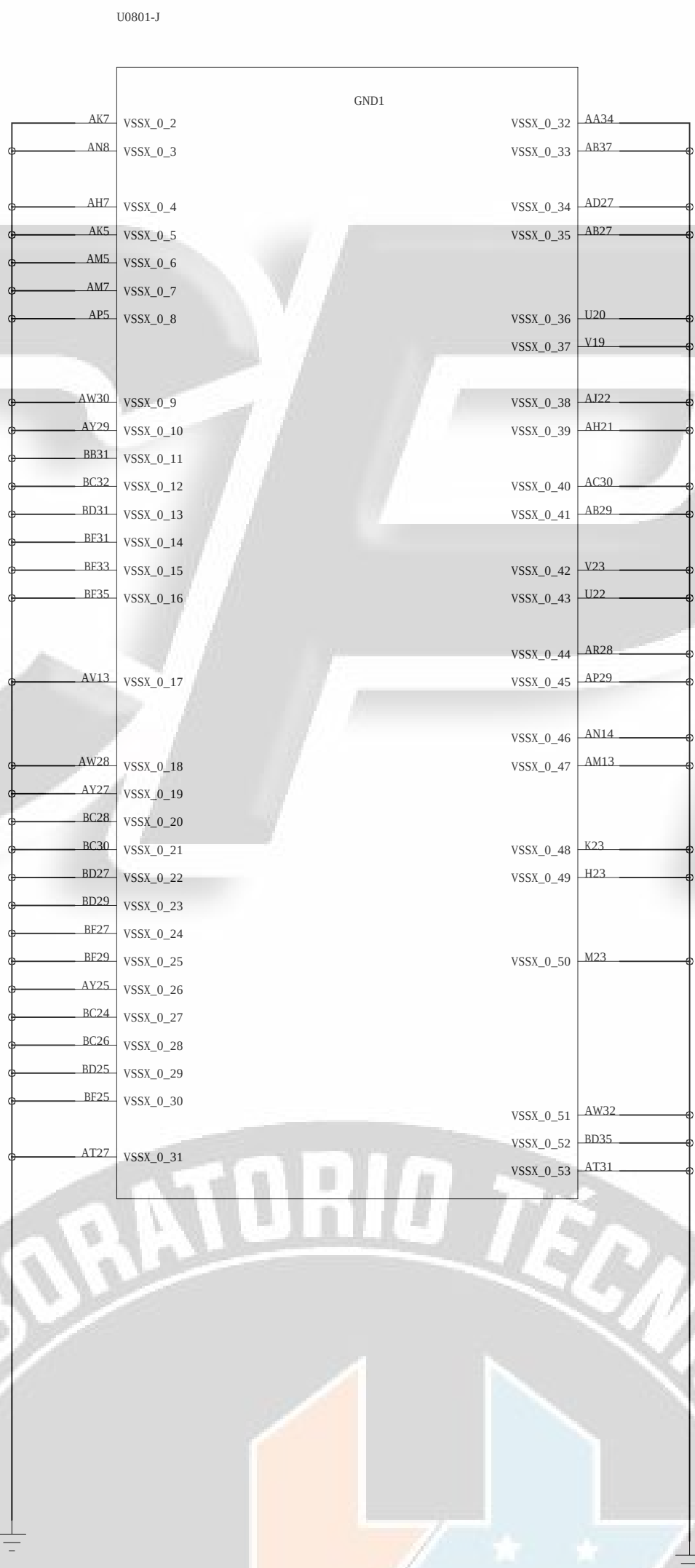
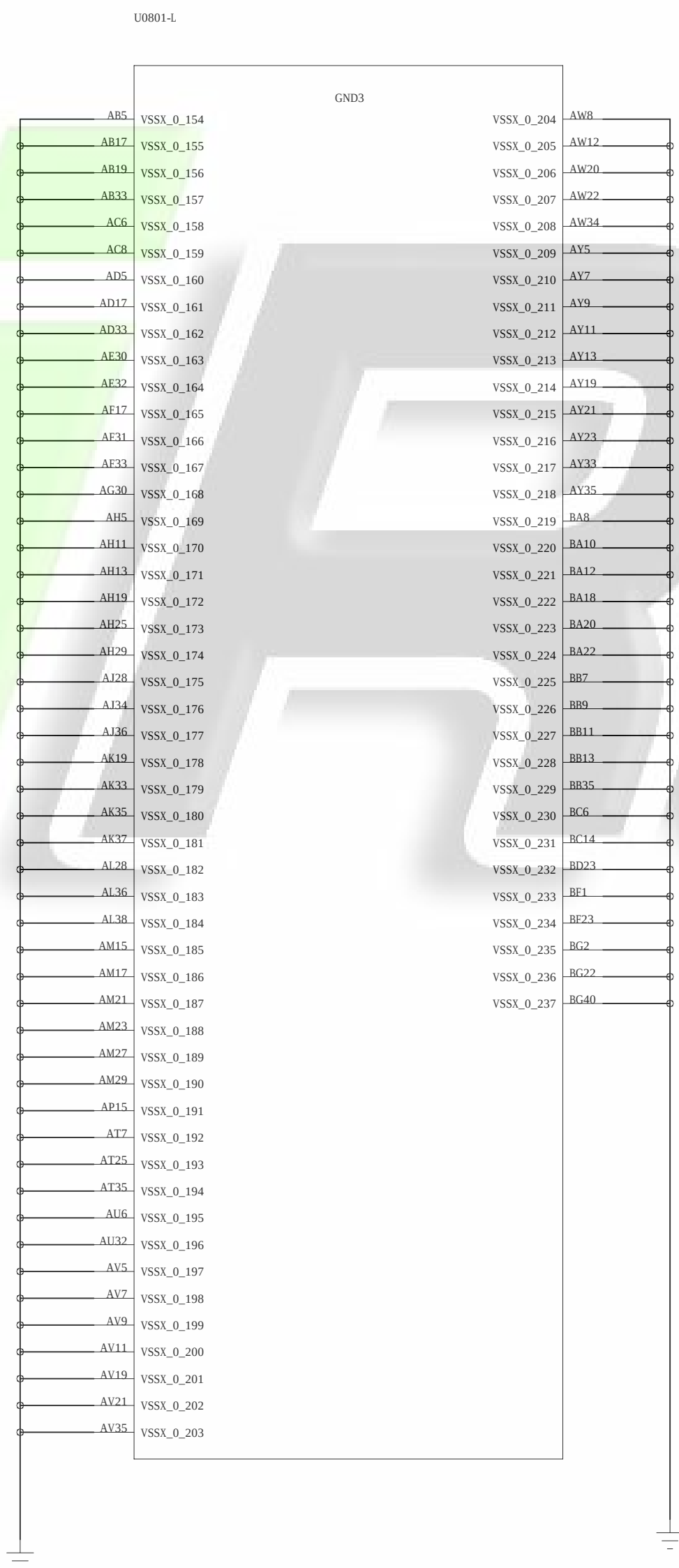
COMPANY: Shanghai Longcheer Technology Co., Ltd			
TITLE: MSM8940-MIPI			
CODE: XX	SIZE: A0	DRAWING NO: XXXX	REV: A

DESIGNER: zhangzhiqiang	DATE: 20160926
CHECKER: yaogang	DATE: 20160926

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

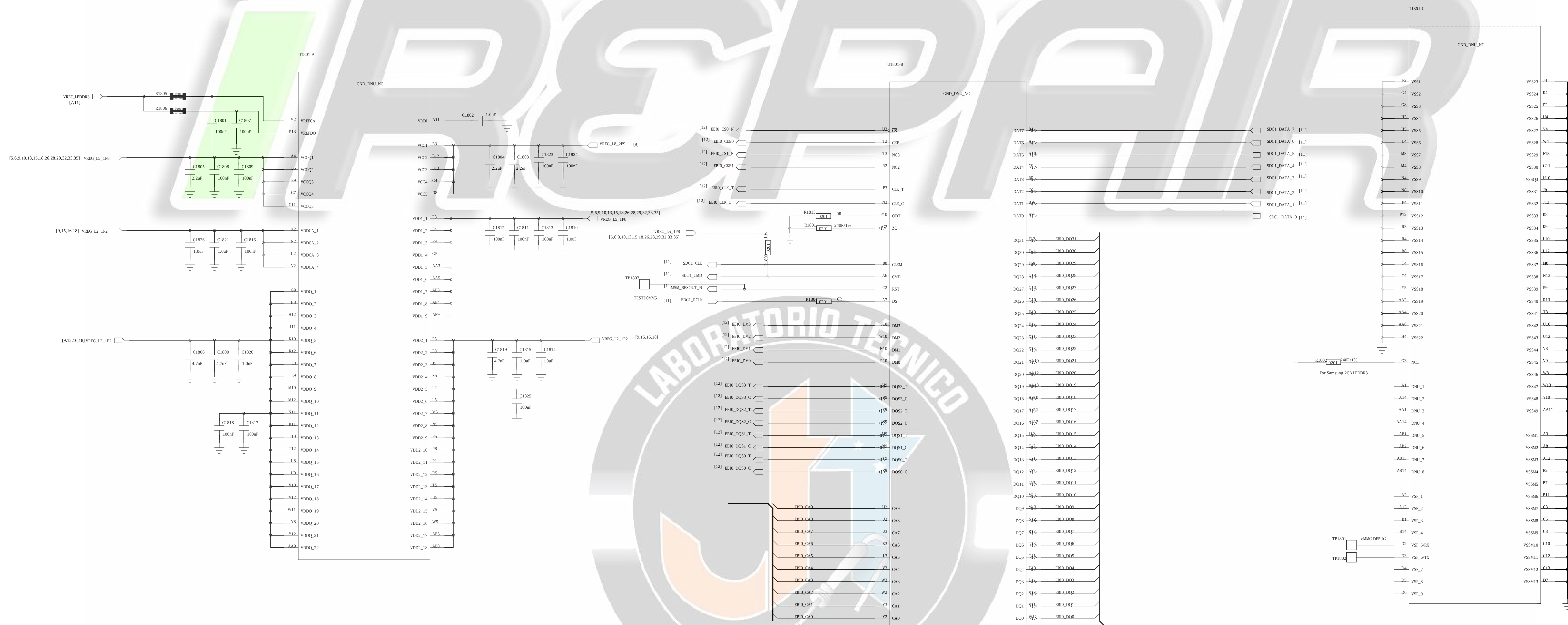


REVISION RECORD			
LTB	ECO NO.	APPROVED	DATE

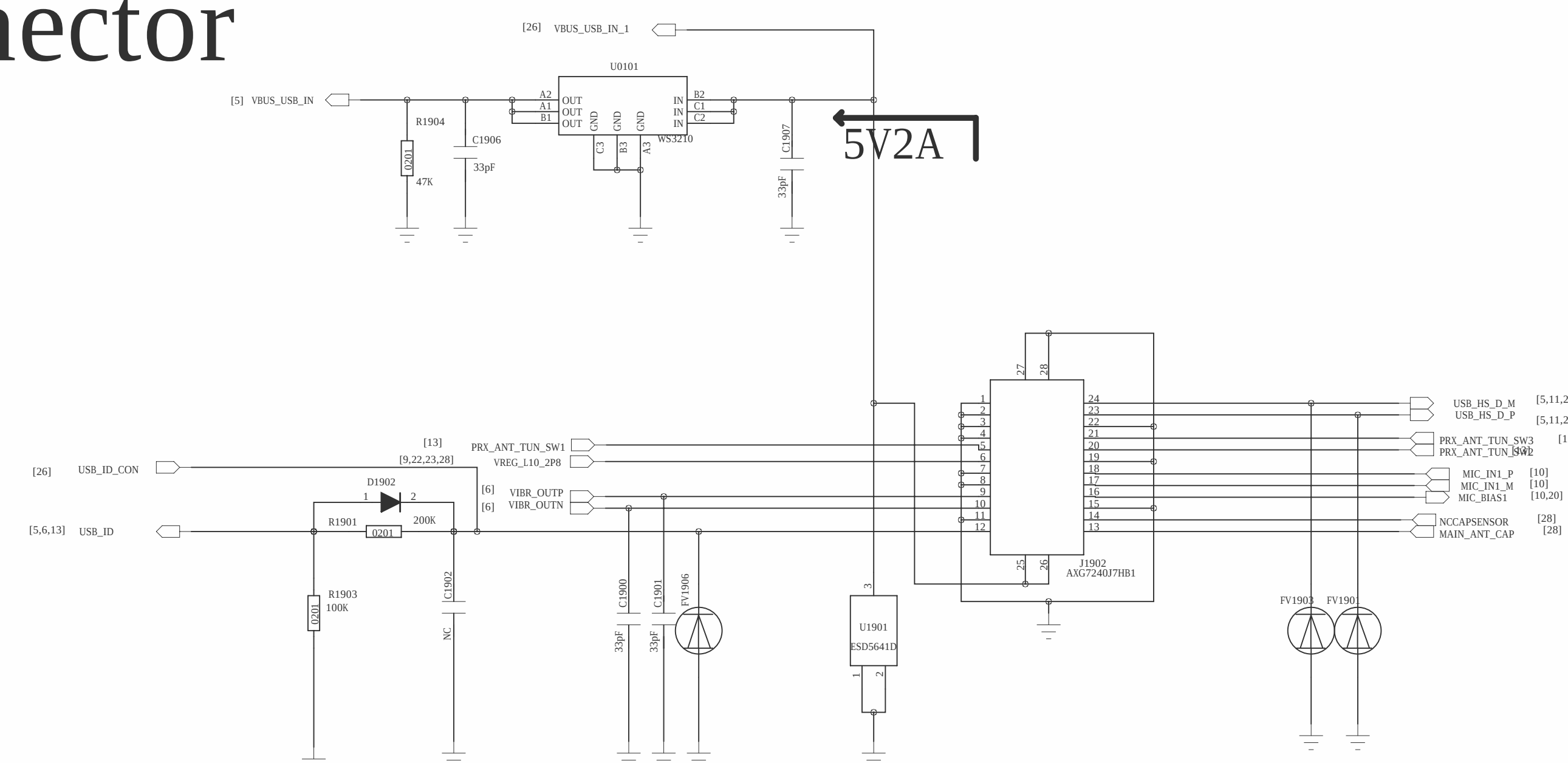


COMPANY: Shanghai Longcheer Technology Co., Ltd			
TITLE: MSM8940-GND			
CODE: XX	SIZE: A0	DRAWING NO: XXXX	REV: A
SHEET: 01 17 OF 36			

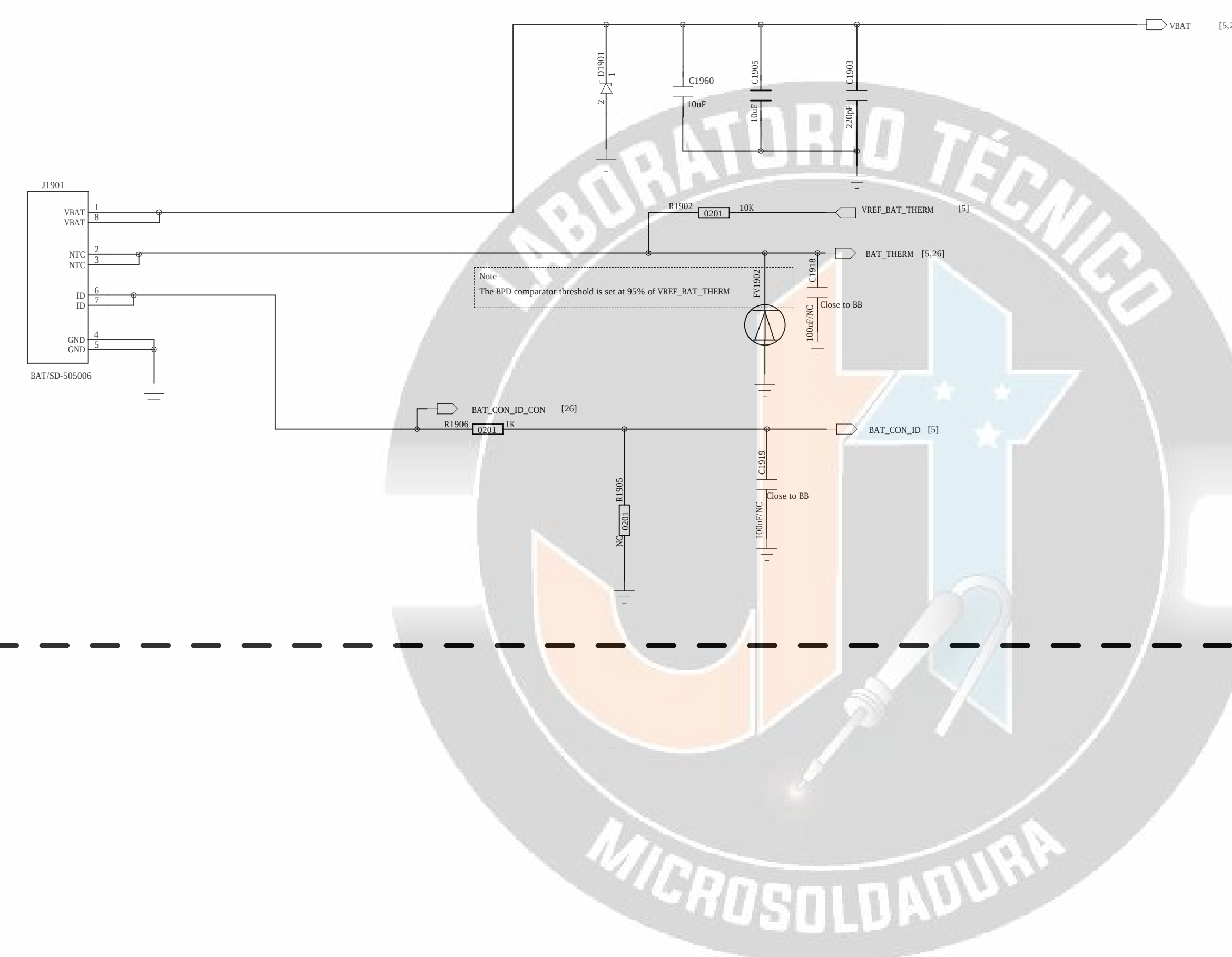
DRAWN: Zhangzhiqiang	DATED: 20160926
CHECKED: yaogang	DATE: 20160926



USB Connector

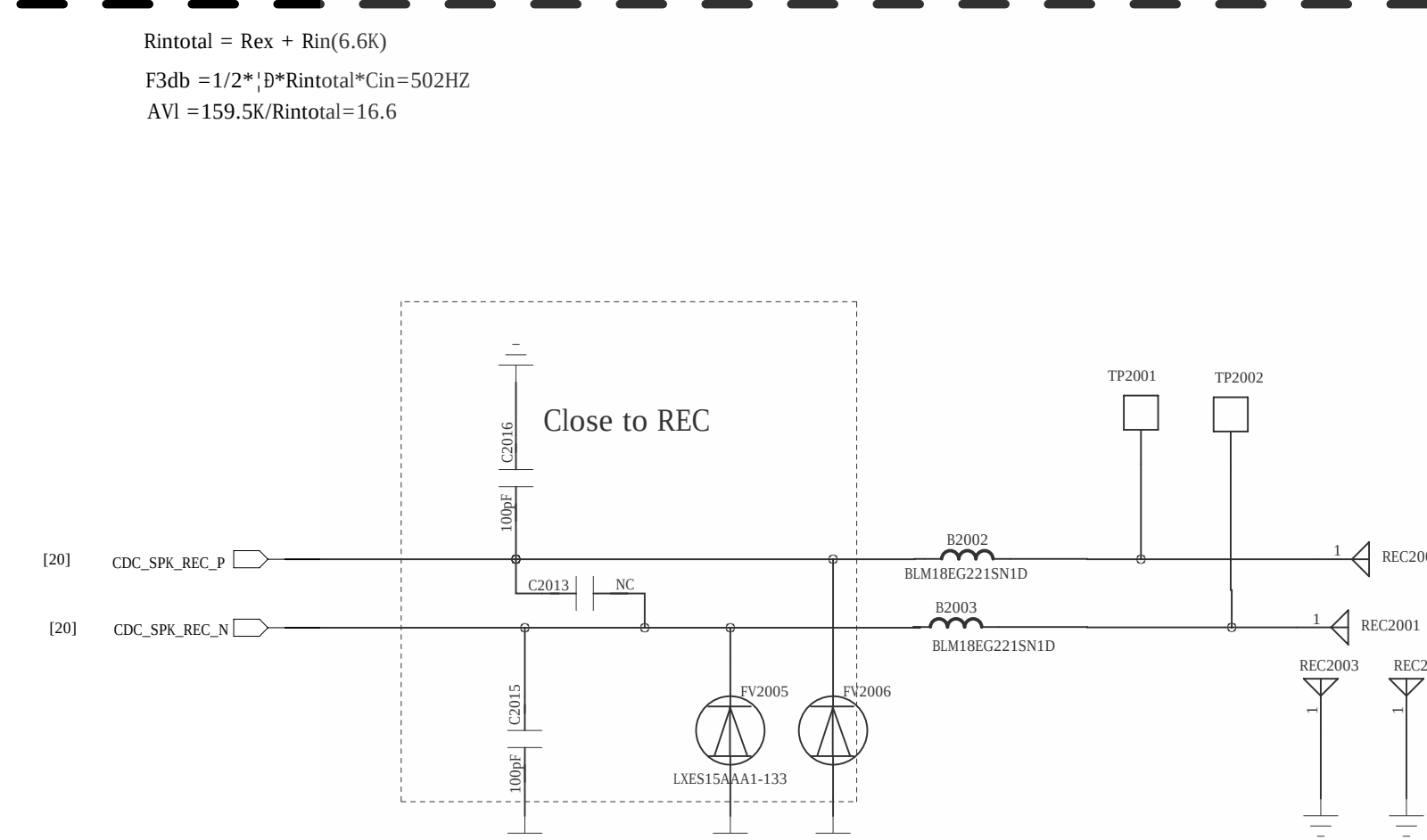


Battery Connector

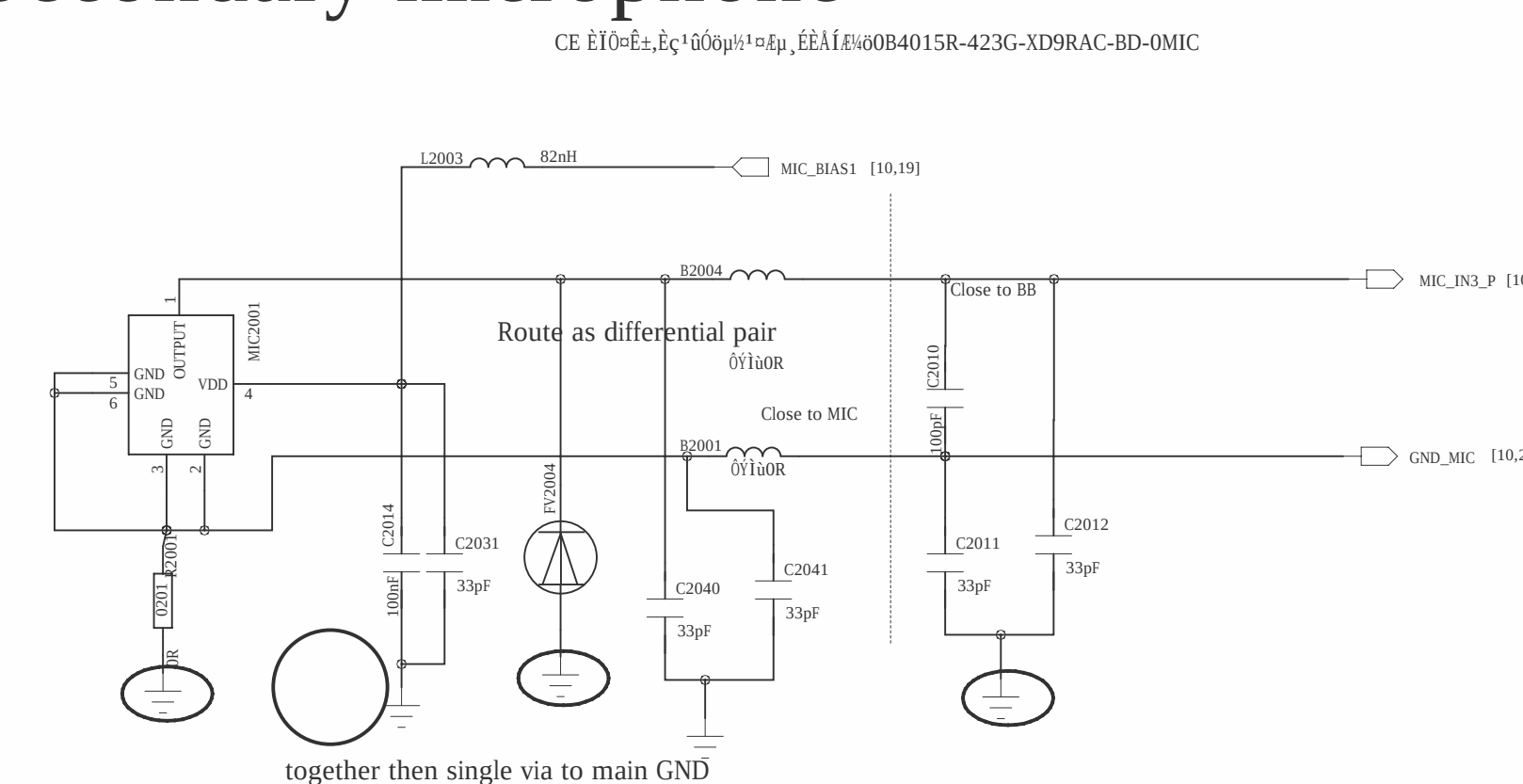


REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

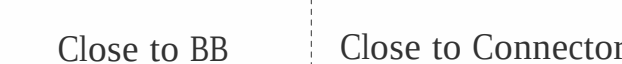
REC



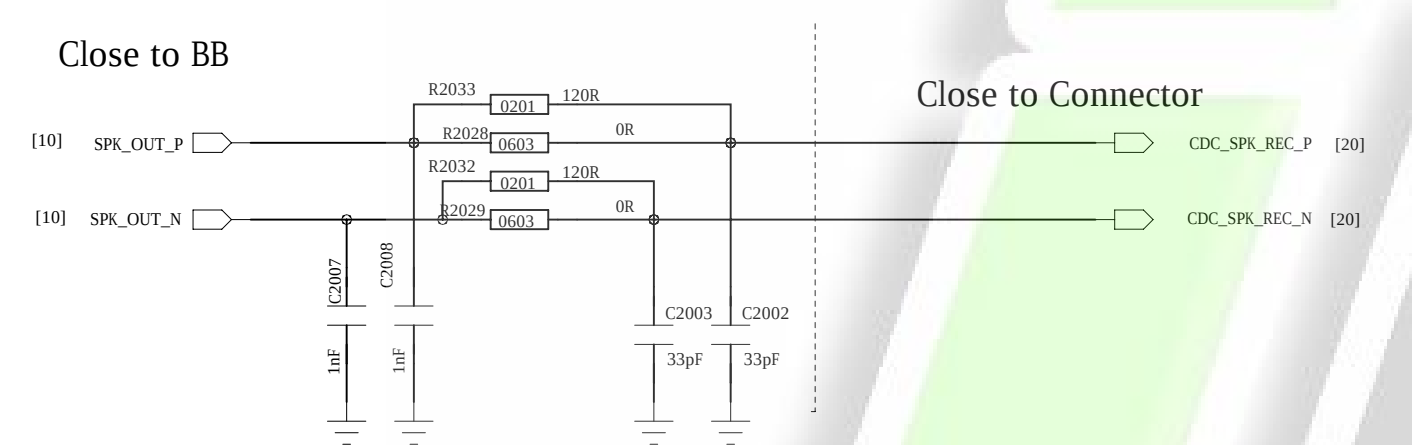
Secondary microphone



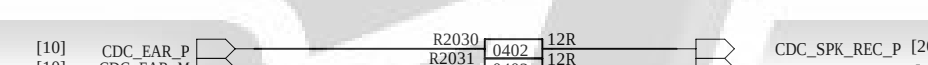
Primary MIC



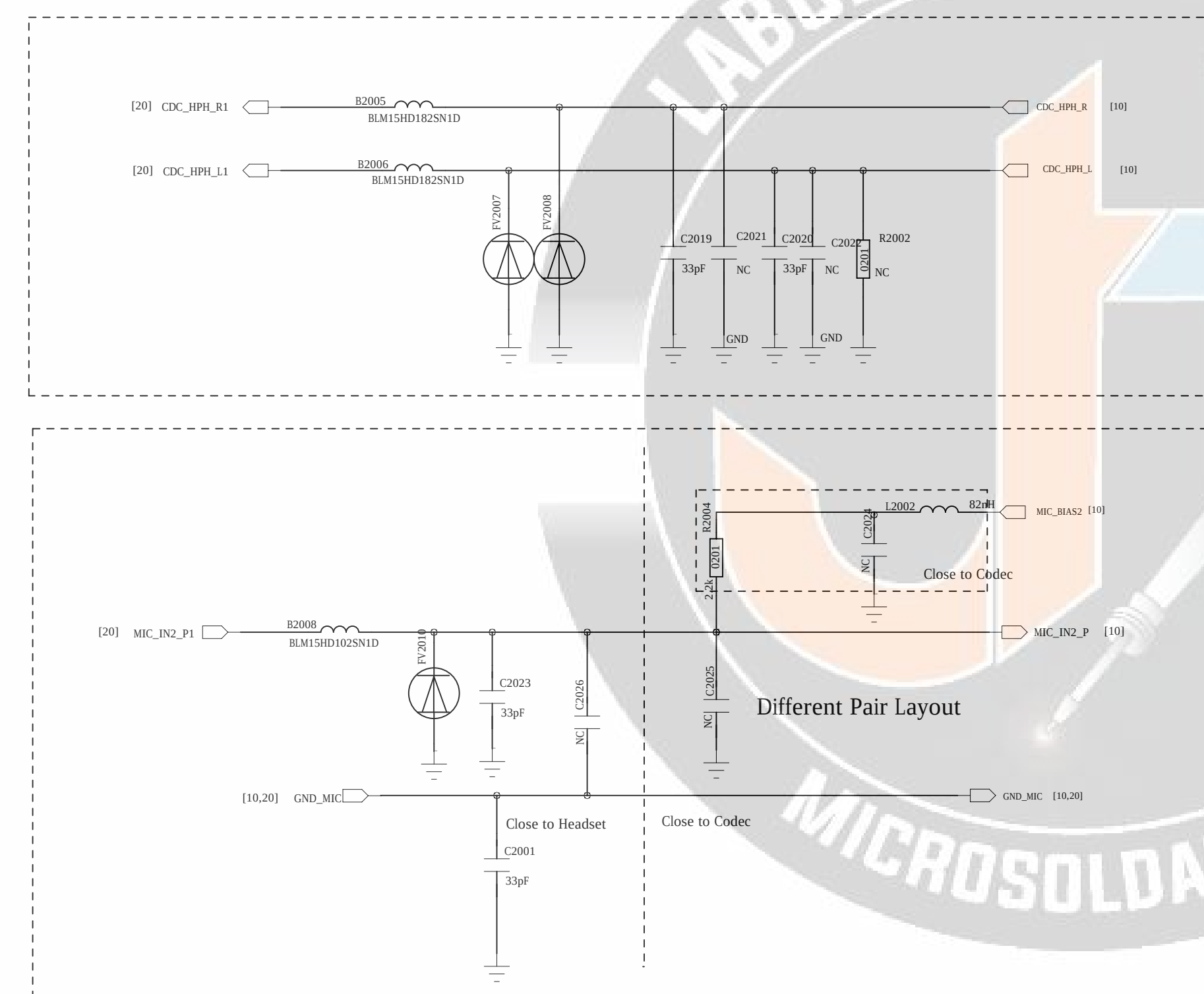
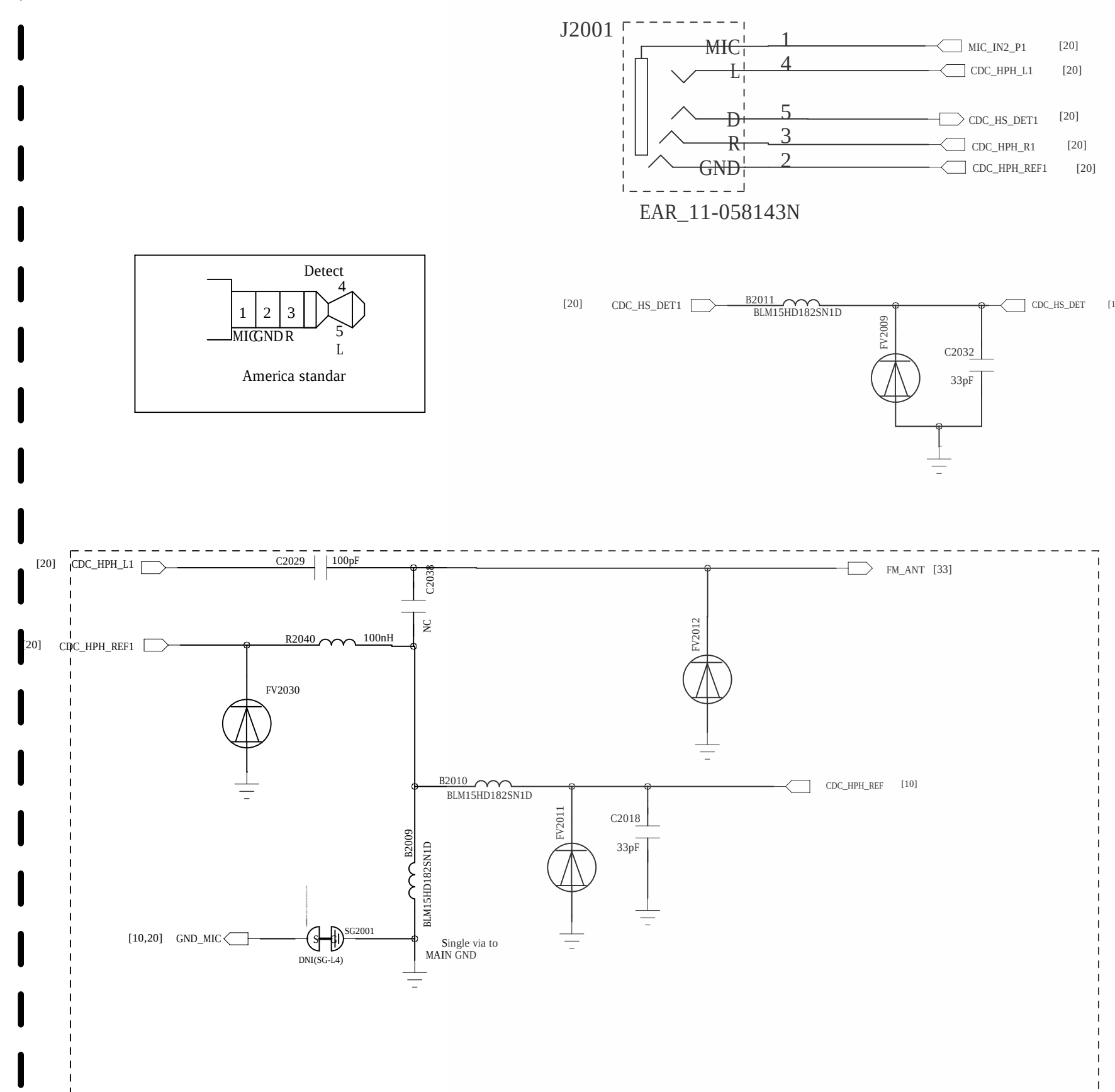
SPK



Audio SW



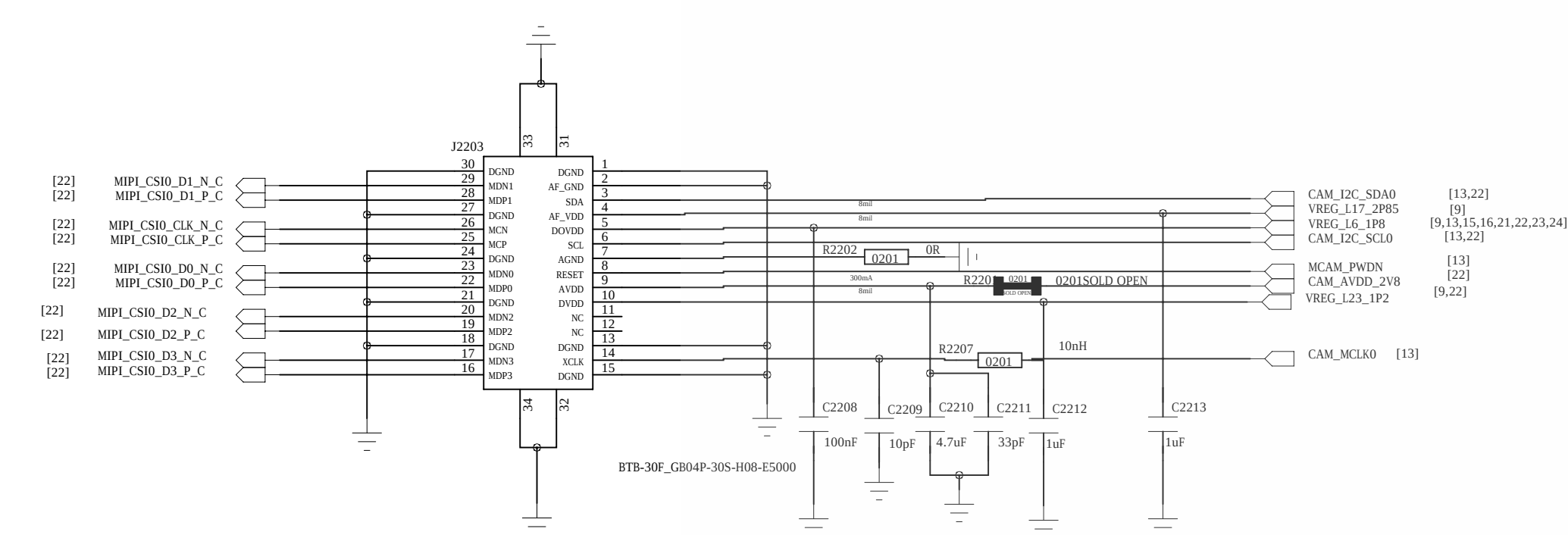
HEADSET



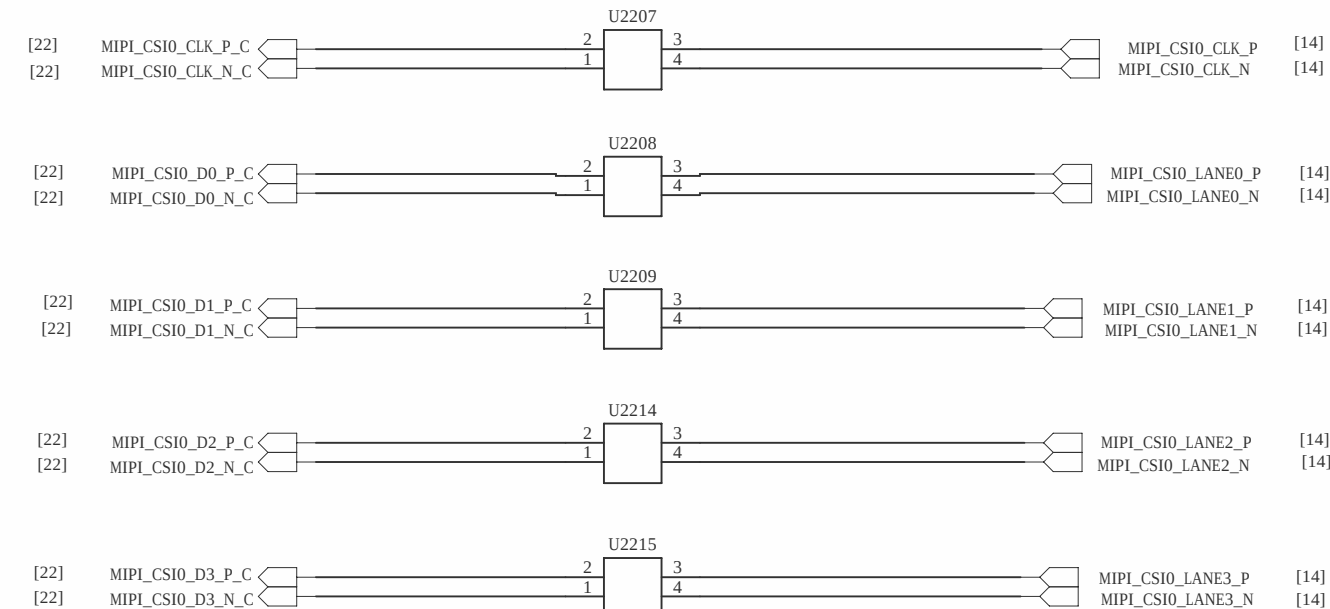
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

Main Camera

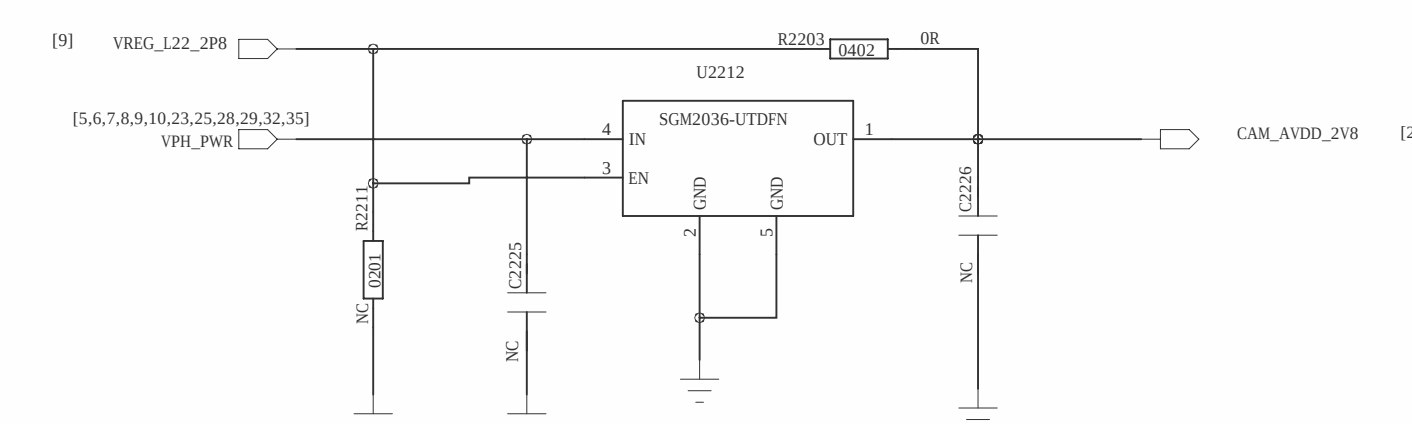
12M



CAD Note
AGND should connect directly to C2432 and then to mainground with dedicated via
AFGND PIN should connect directly to C2434 and then to mainground with dedicated via

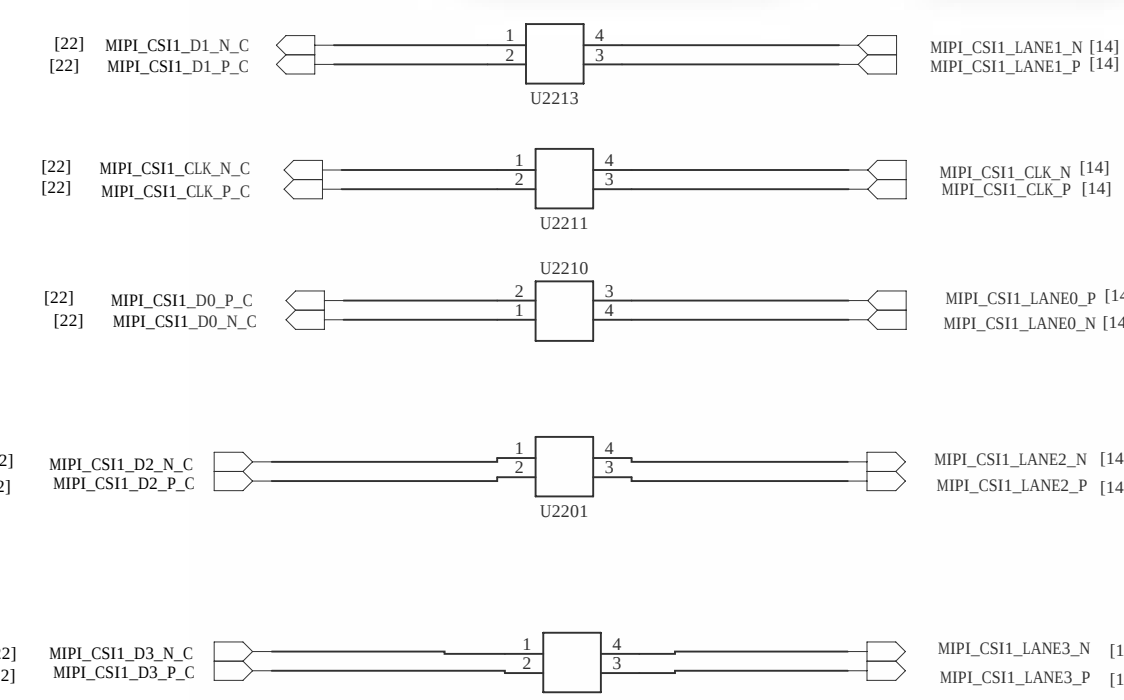
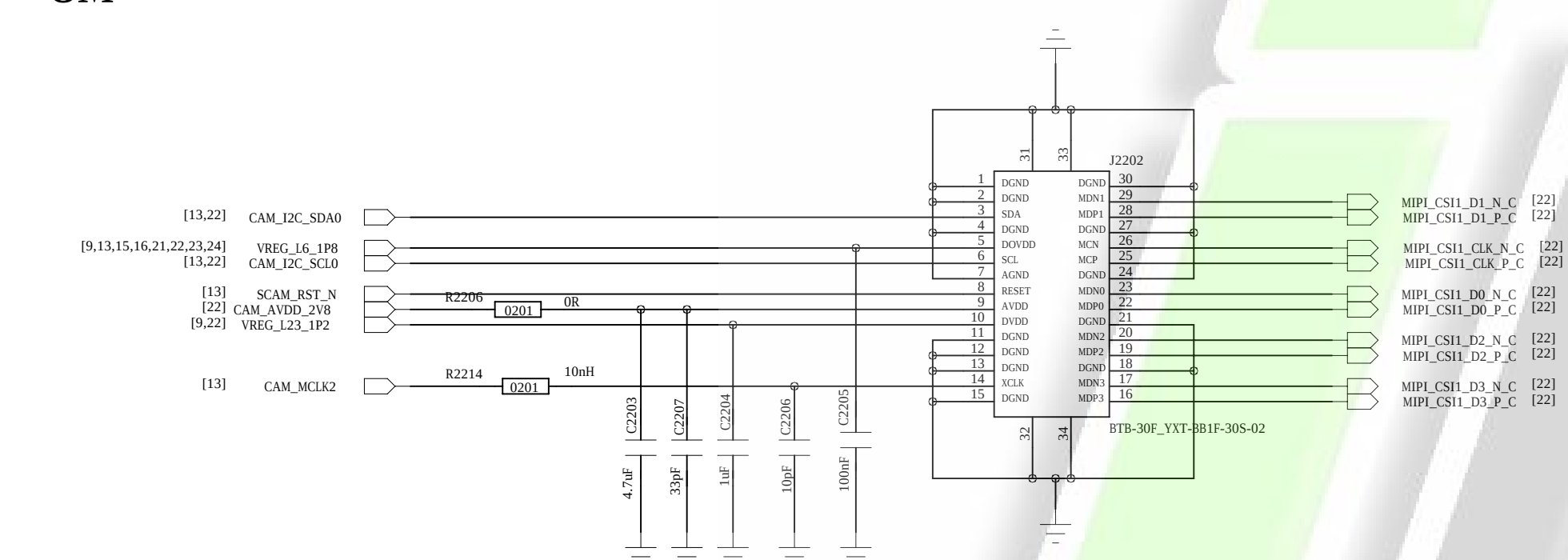


LDO

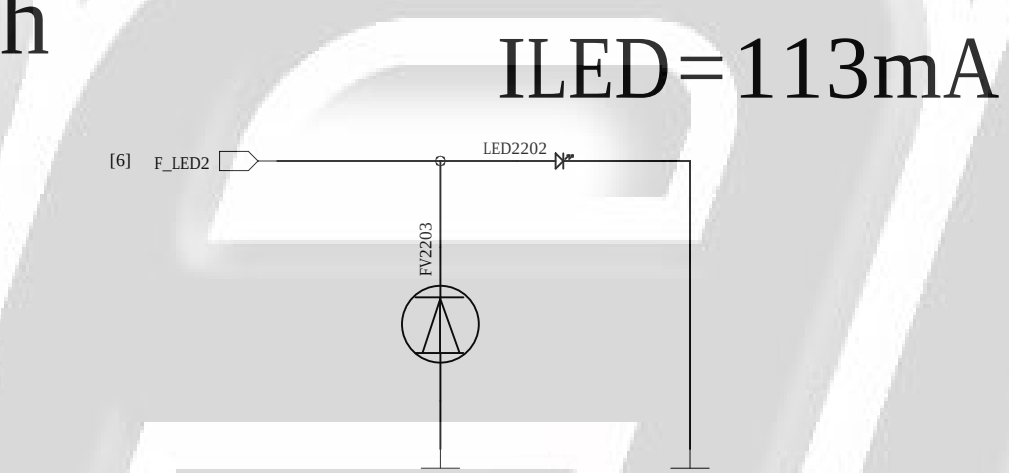


Front Camera

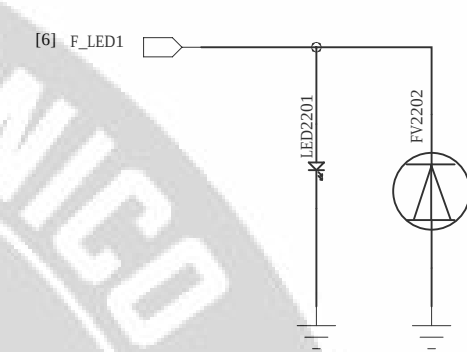
8M



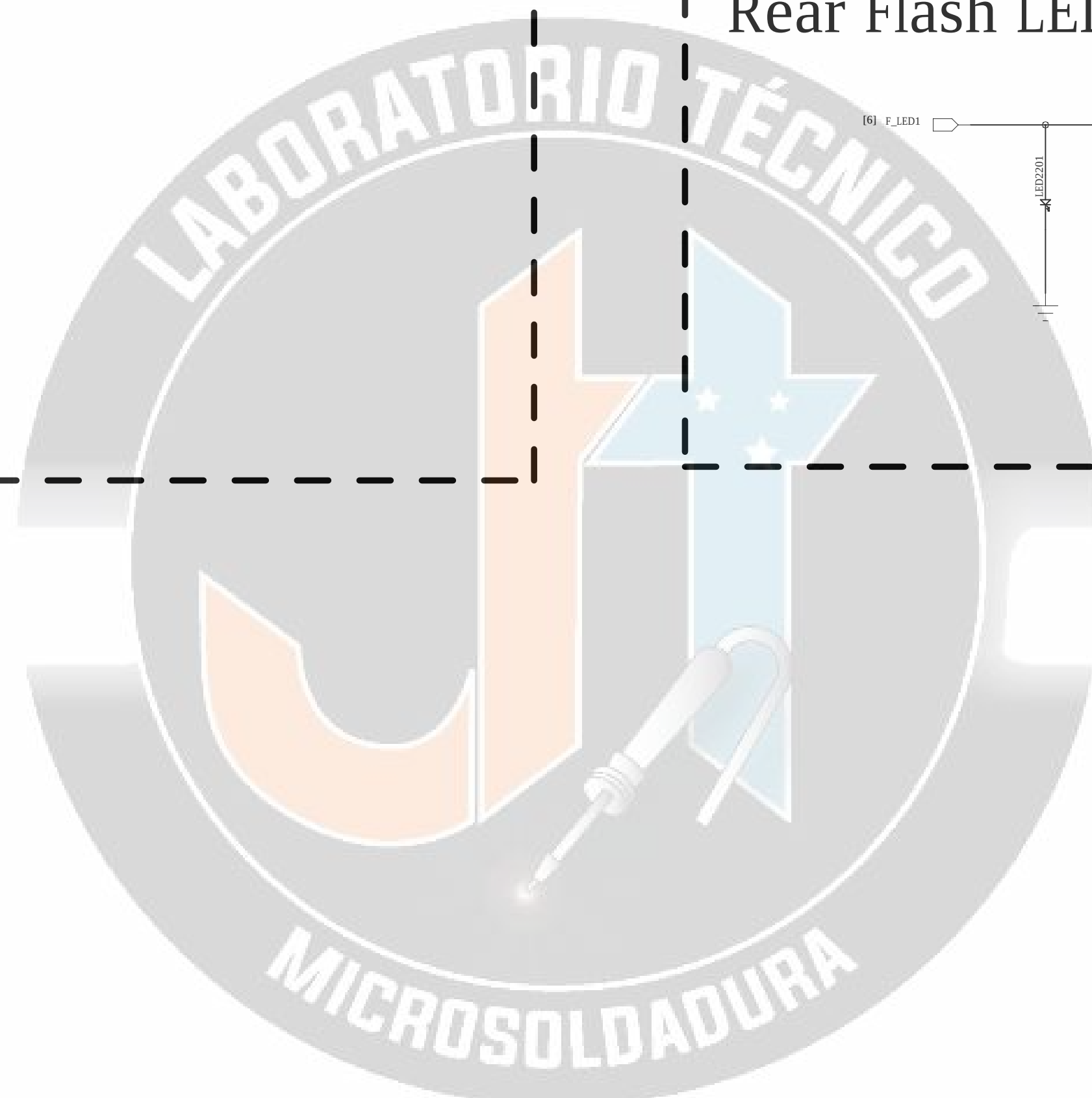
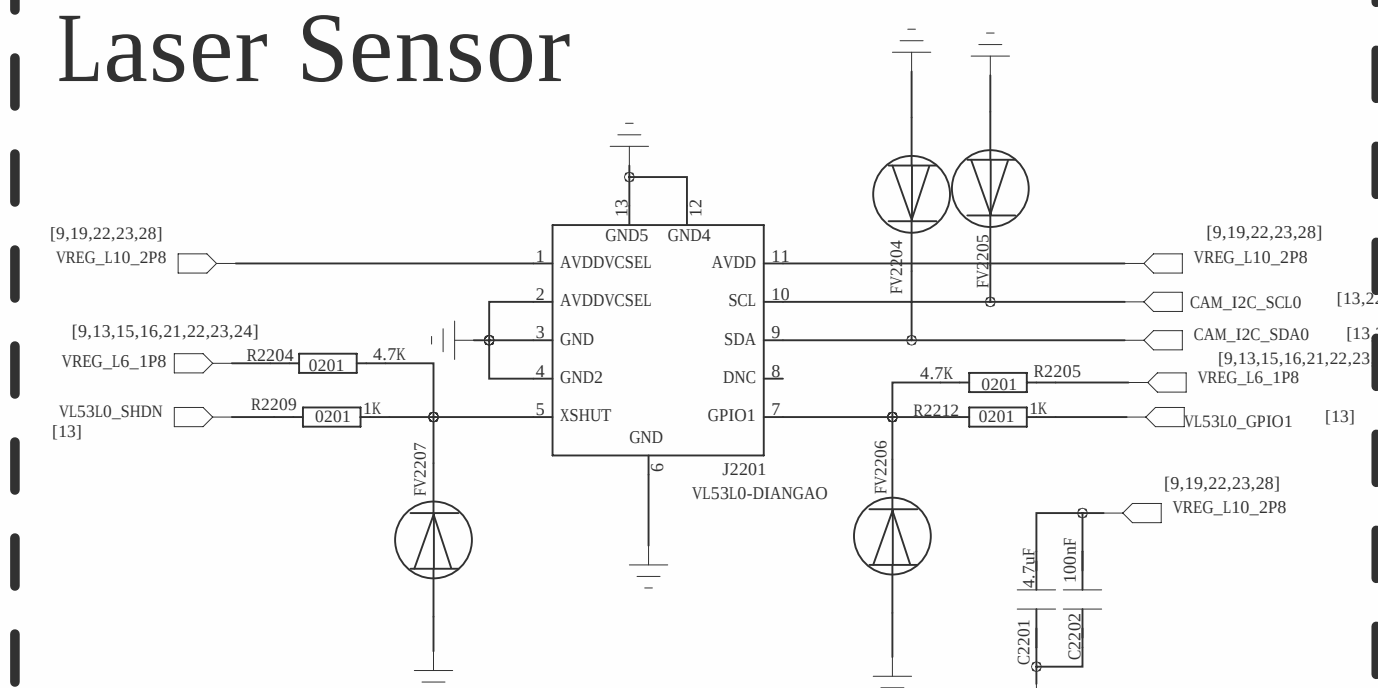
Front Flash



Rear Flash LED



Laser Sensor

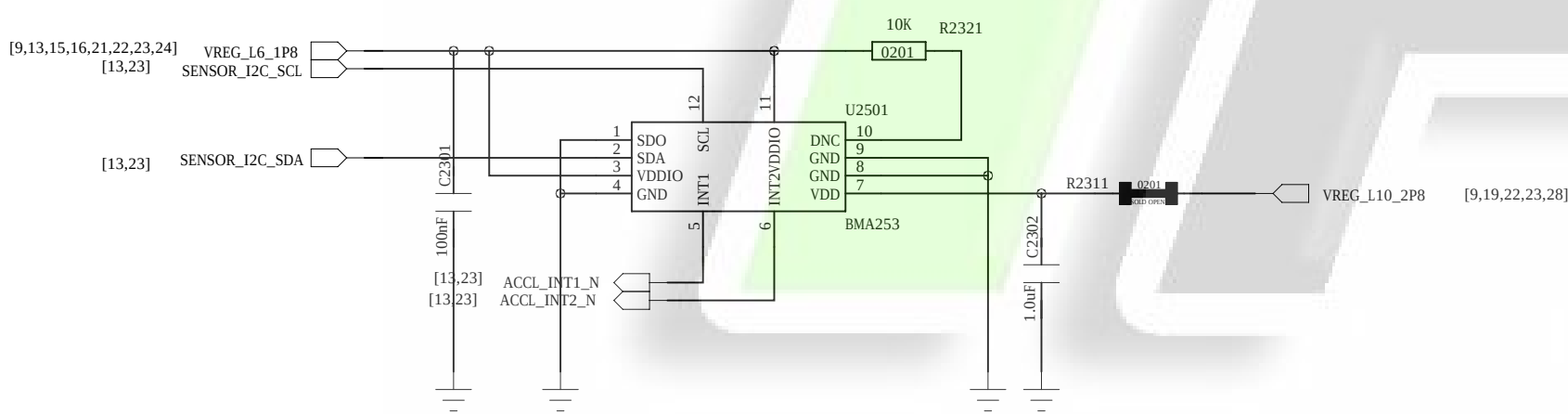


REVISION RECORD			
VER	ECO NO.	APPROVED	DATE

Accelerometer Sensor

8bit(BMA223) compatible 12bit(BMA253)

ADDR 0011000b(0x18) SDO=0

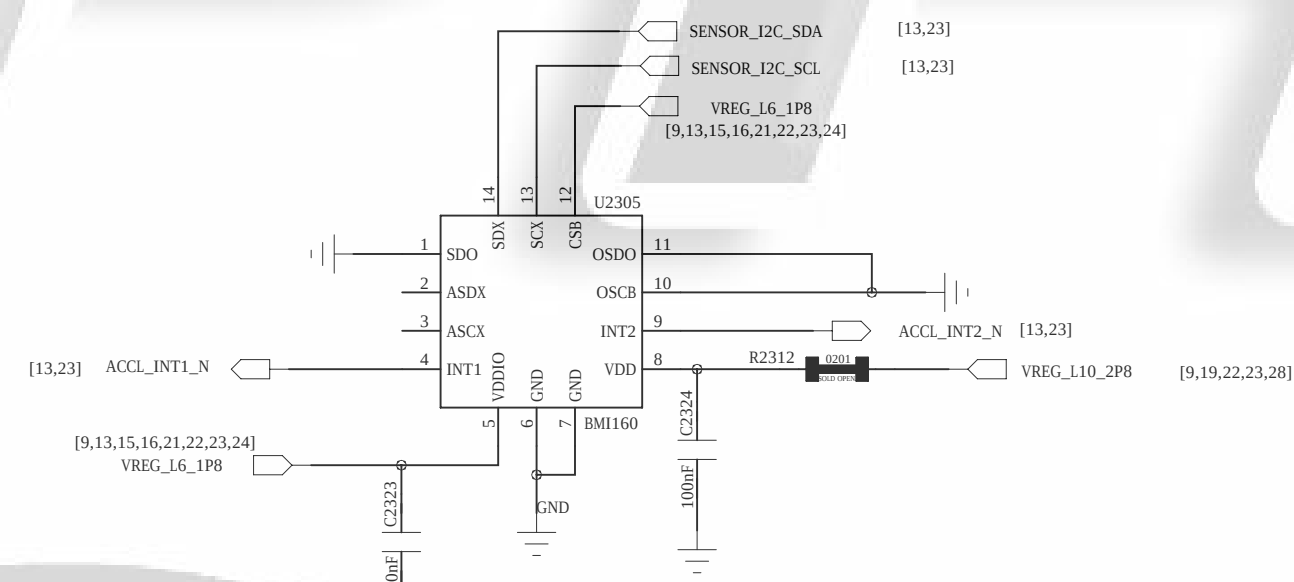


G+Gyro-Sensor

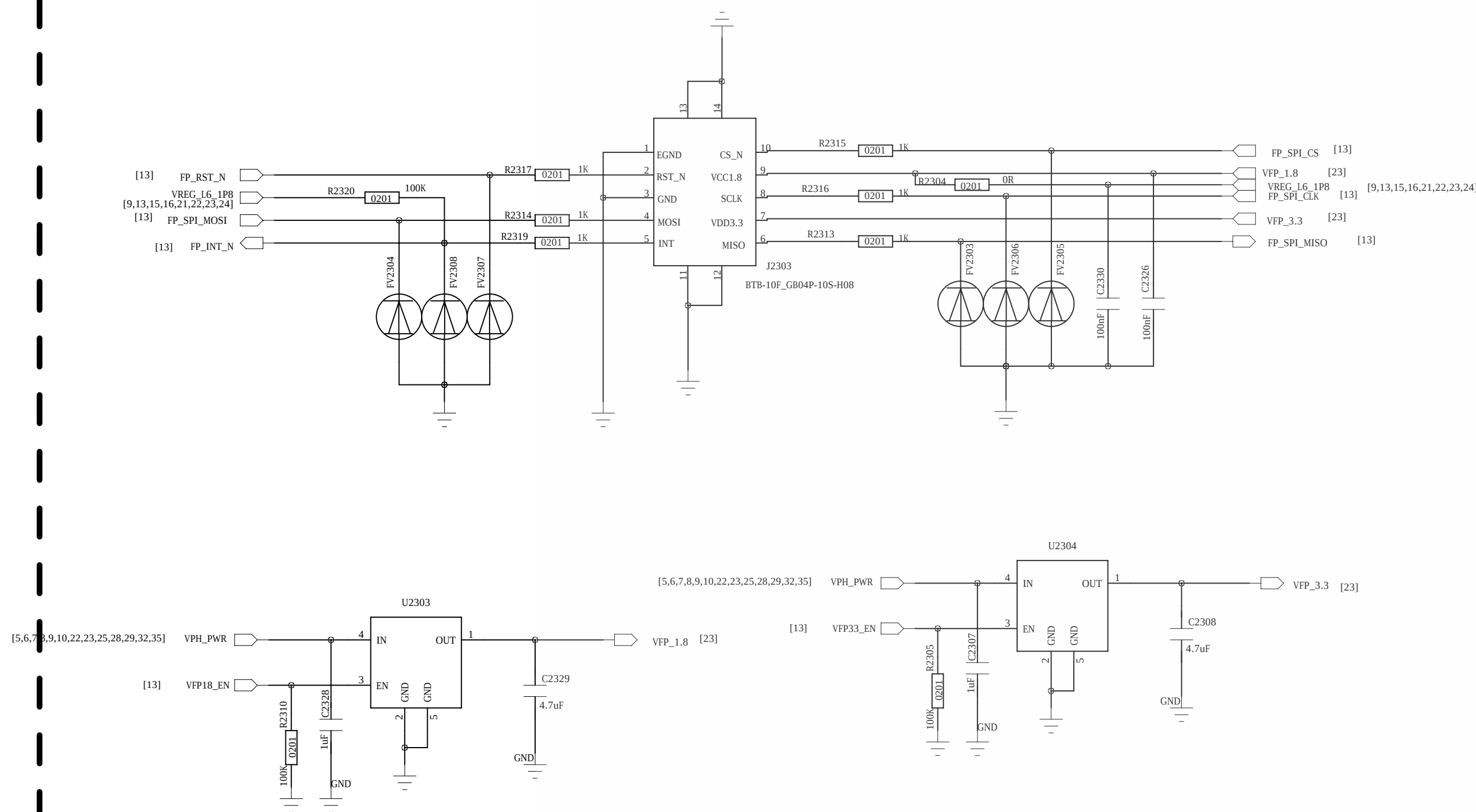
BMI120 I2C address

SDO=0; 0x68

SDO=1; 0x69



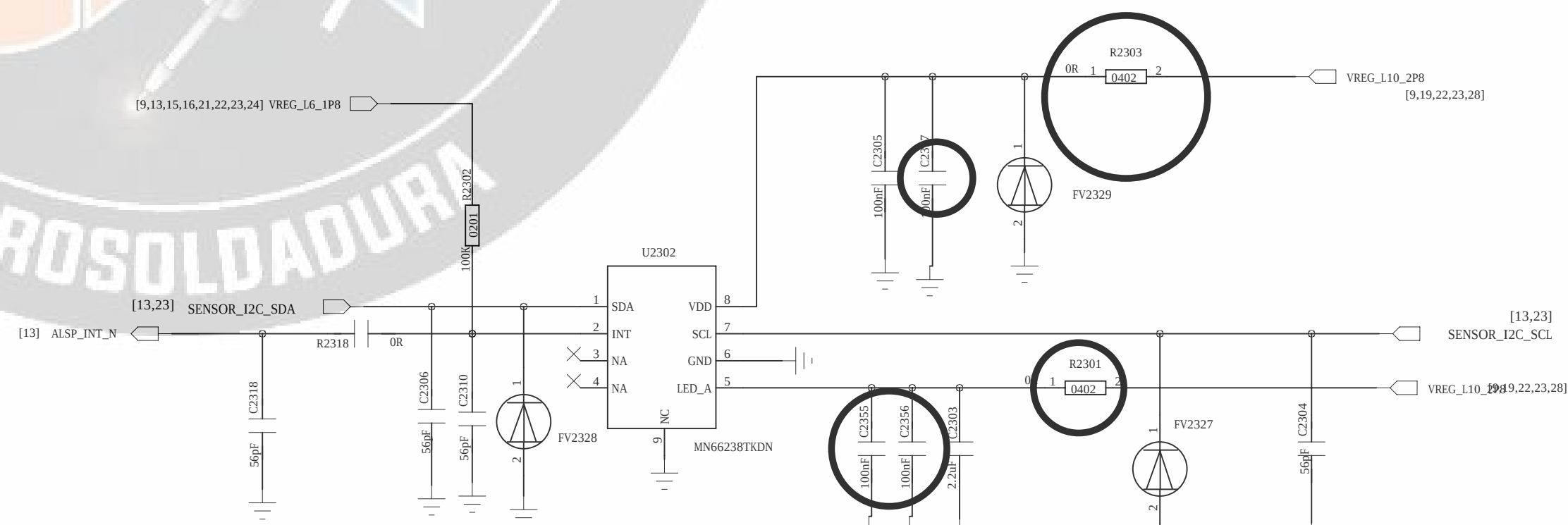
FingerPrint-Sensor



PS Sensor

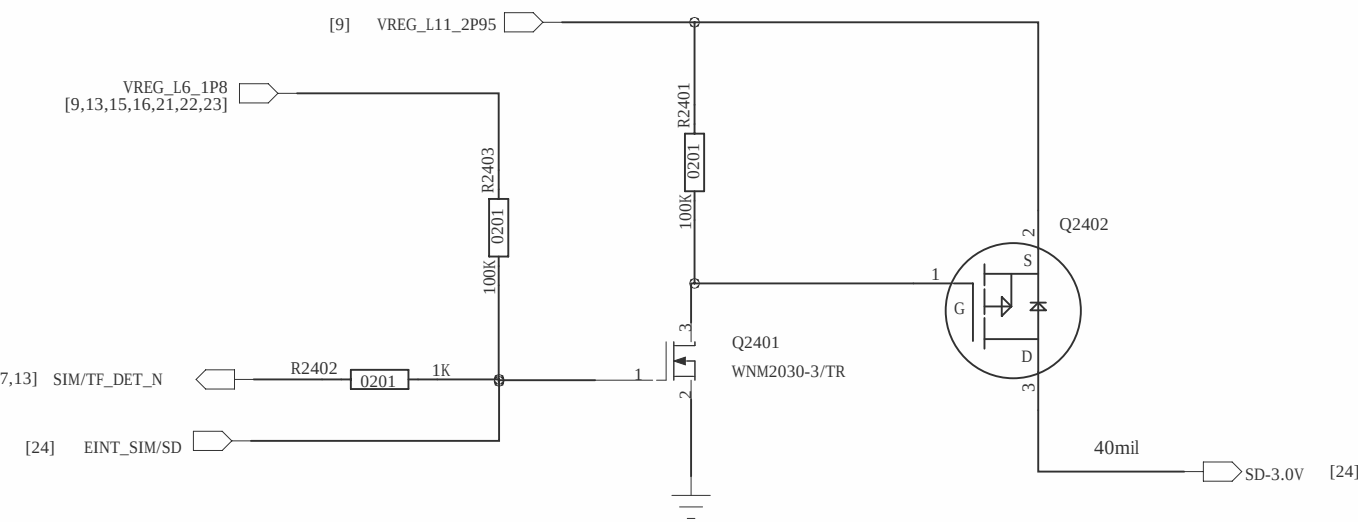
EPL2590KTWP

I2C address: 0xA6H Write 0xA7H Read

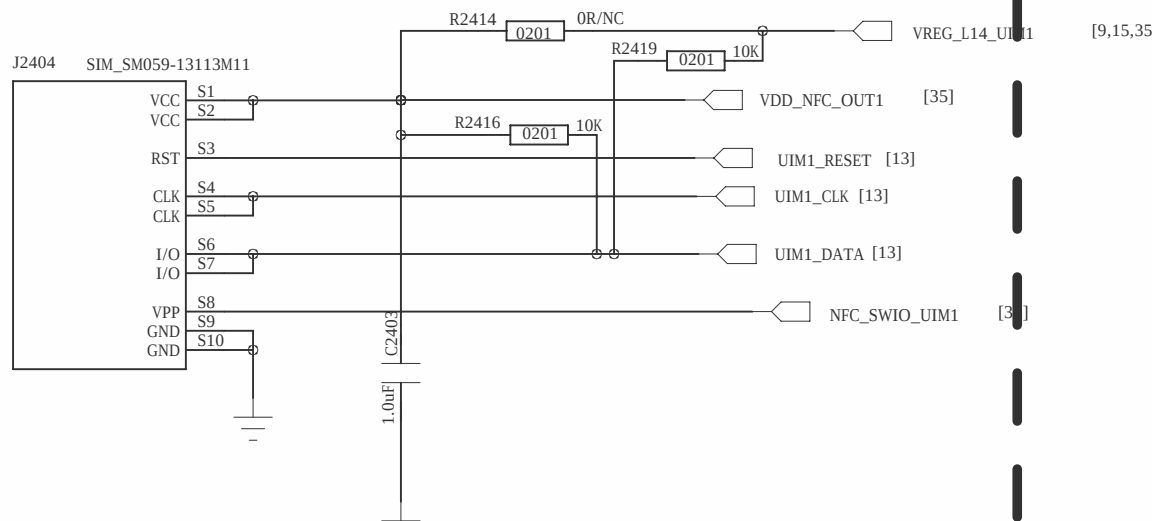


REVISION RECORD			
VER	ECO NO.	APPROVED	DATE

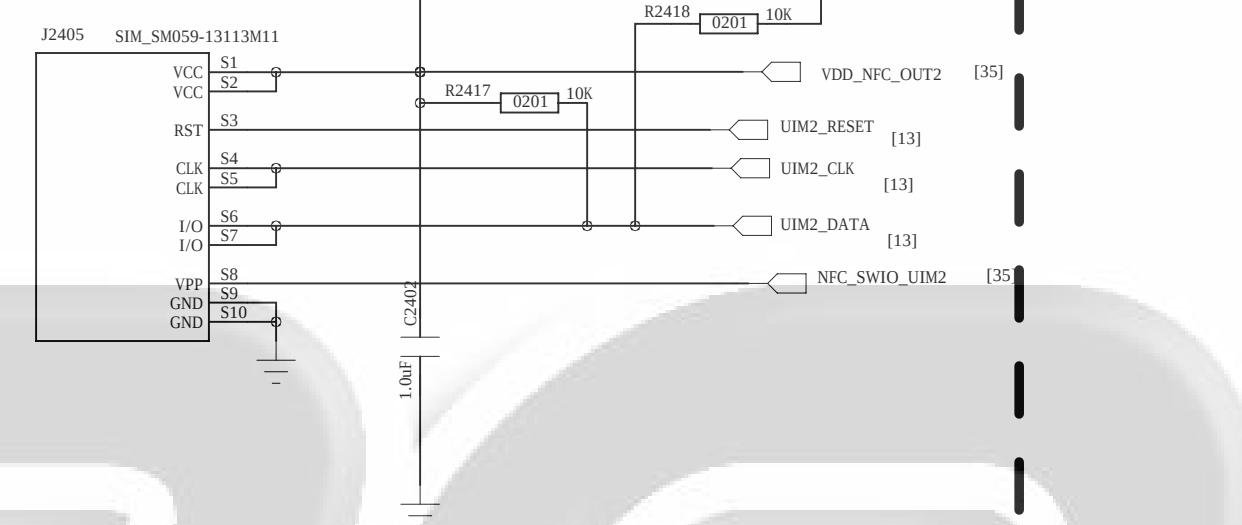
From P3585



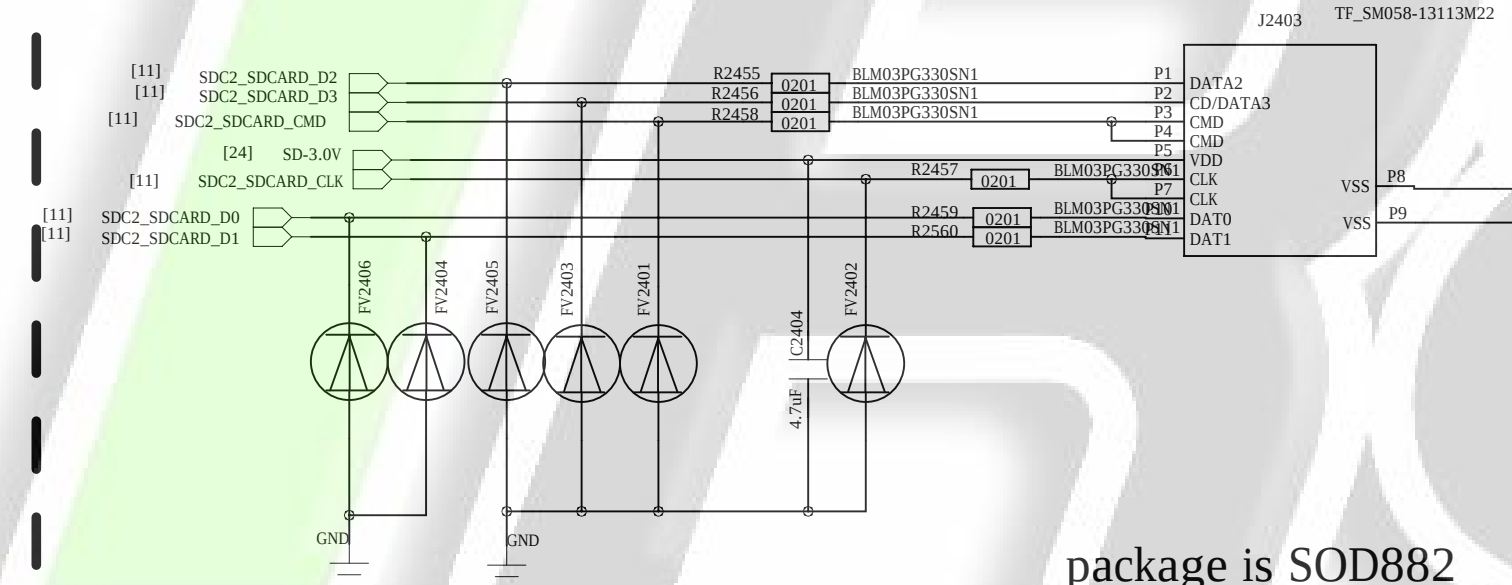
SIM1



SIM2

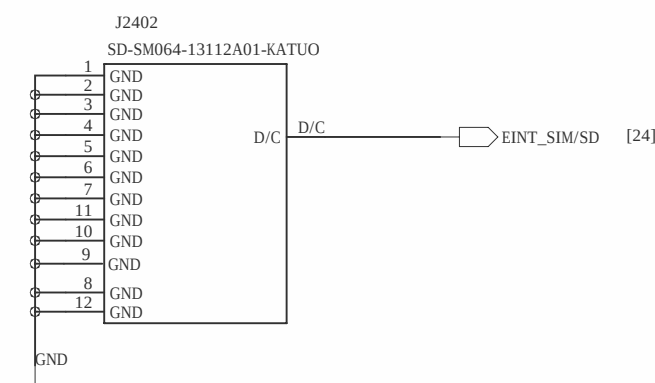


SD CARD

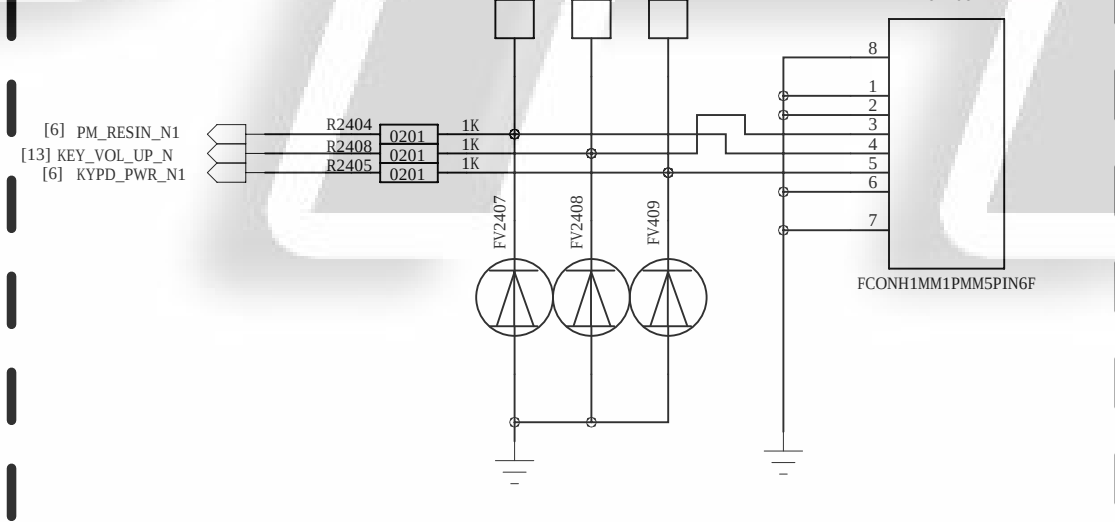


package is SOD882

Card shielding



KEY



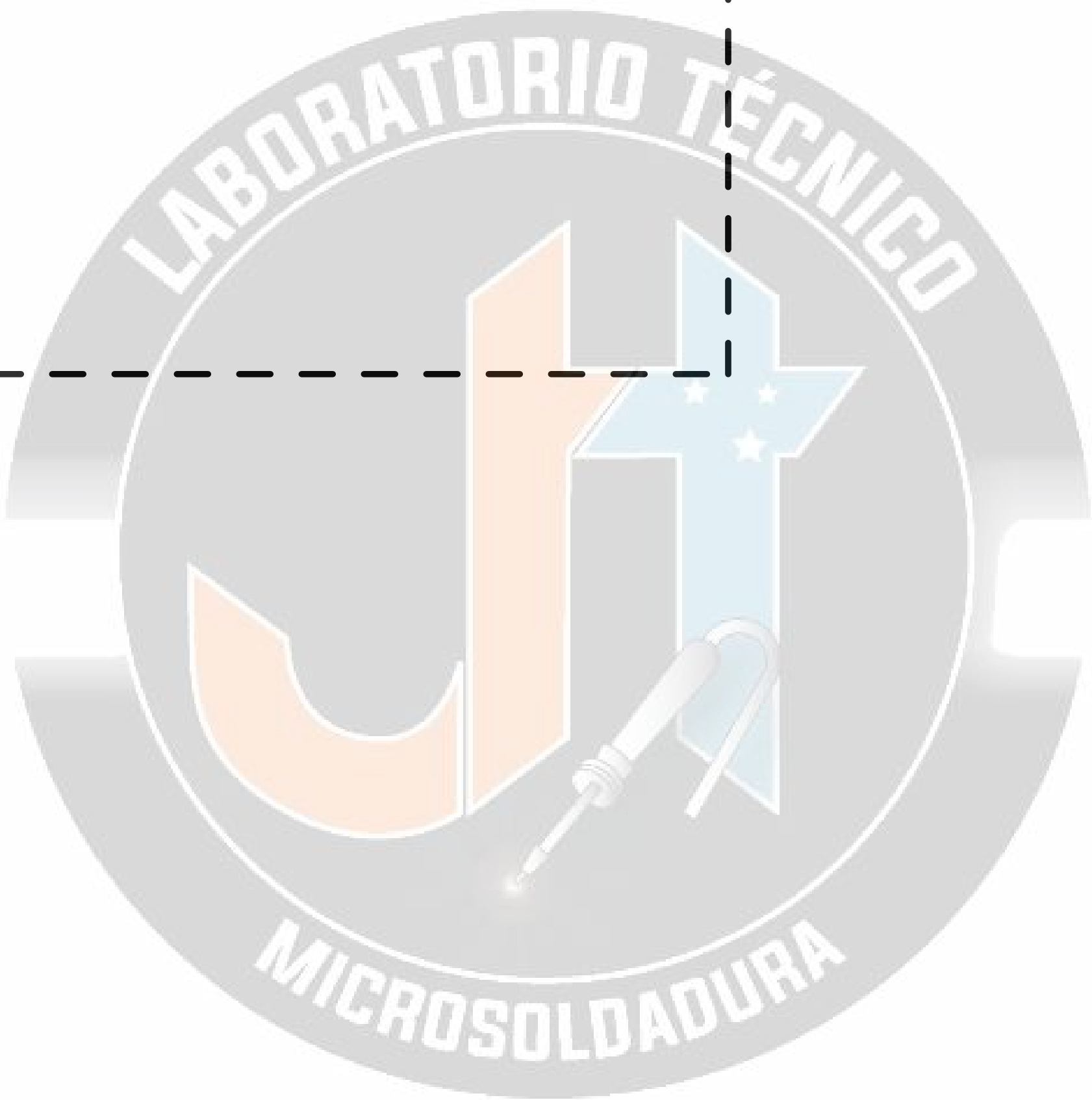
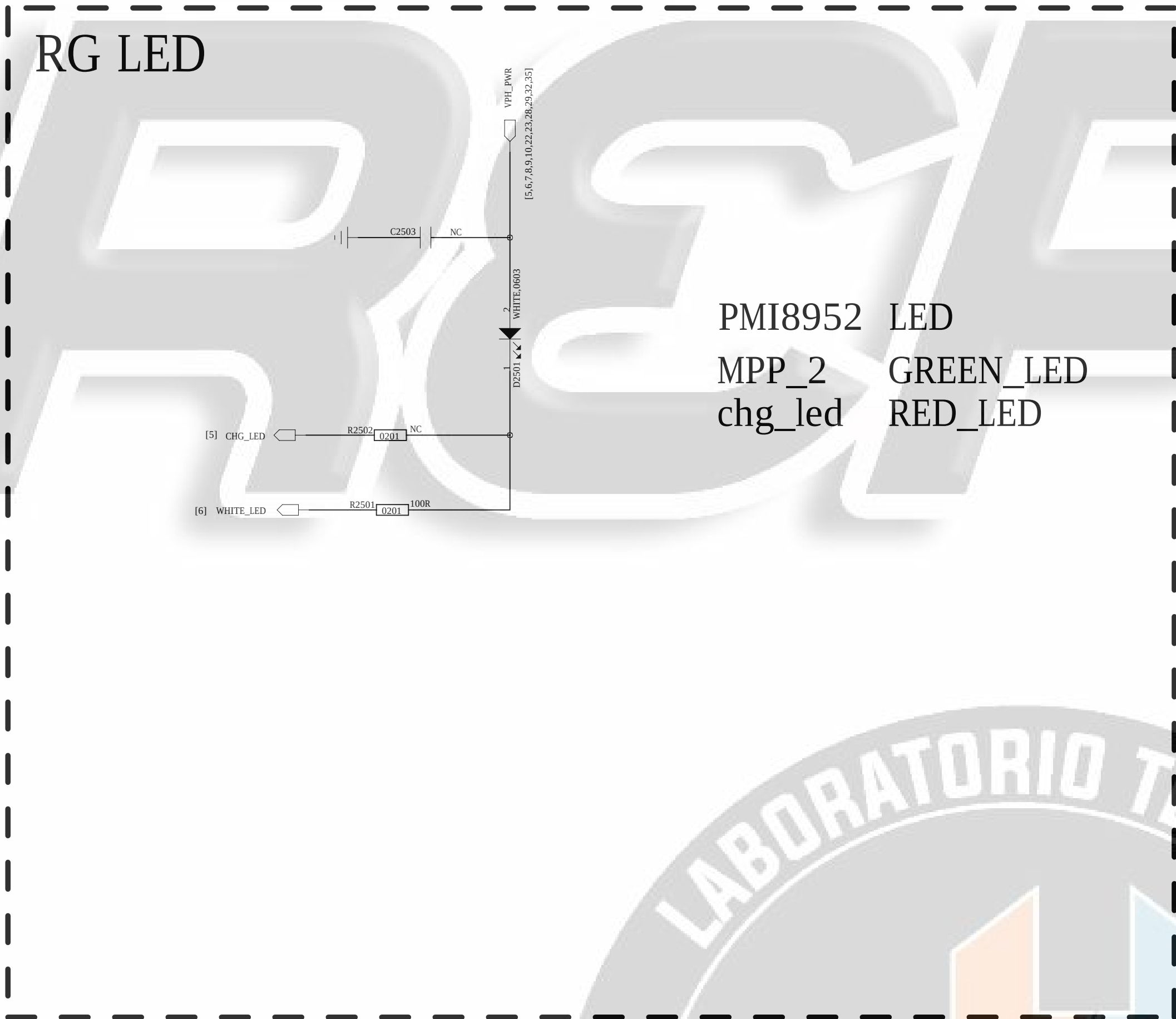
Power Key / Key Pad
DO NOT put pull-up resistor on PWRKEY



COMPANY: Shanghai Longcheer Technology Co., Ltd			
TITLE: SIM_TF CARD			
CODE	SIZE	DRAWING NO.	REV
XX	A0	XXXX	A

DESIGN: zhangzhiqiang	DATED: 20160926
DRAWN: yaogang	DATED: 20160926

REVISION RECORD			
LTB	ECO NO.	APPROVED	DATE

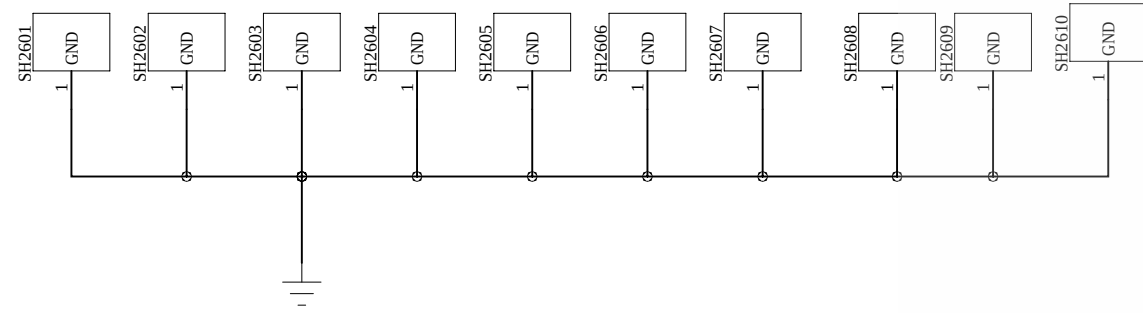


COMPANY: Shanghai Longcheer Technology Co., Ltd			
TITLE: RG_LED/MOT			
CODE:	SIZE:	DRAWING NO:	REV:
XX	A0	XXXX	A

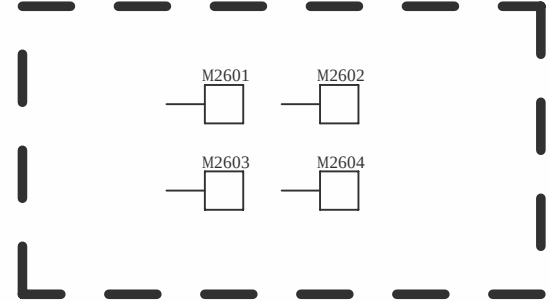
DESIGNED:	DATE:
zhangzhiqiang	20160926
CHECKED:	DATE:
vaogang	20160926

SCALE: <Scale>	SHEET: OF 25 OF 36
----------------	--------------------

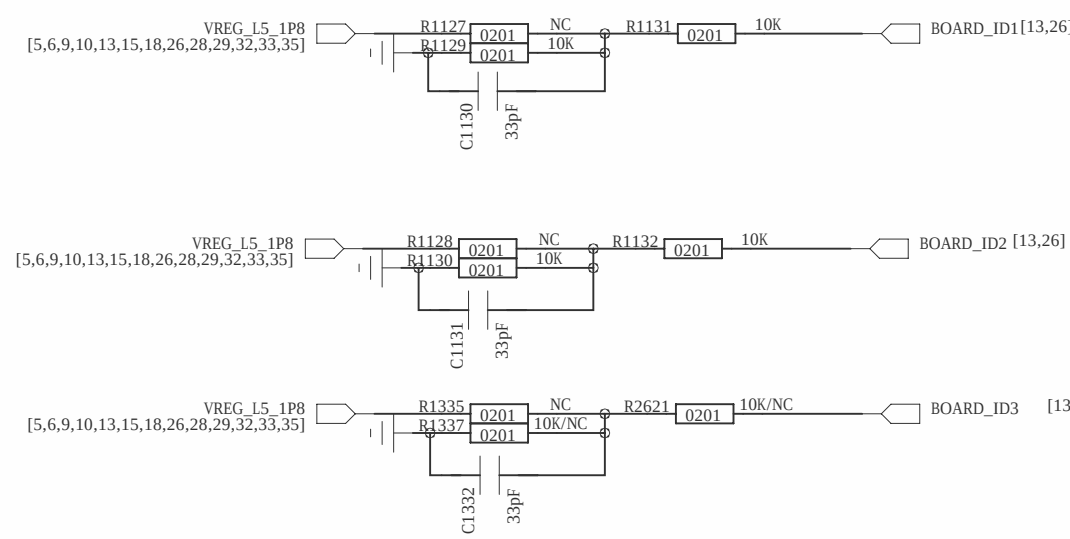
REVISION RECORD			
VER	ECO NO.	APPROVED	DATE



Mark Point



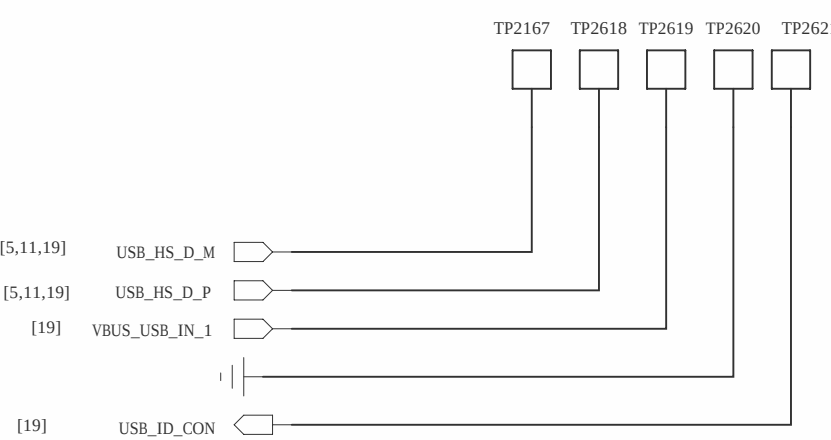
board ID



board ID

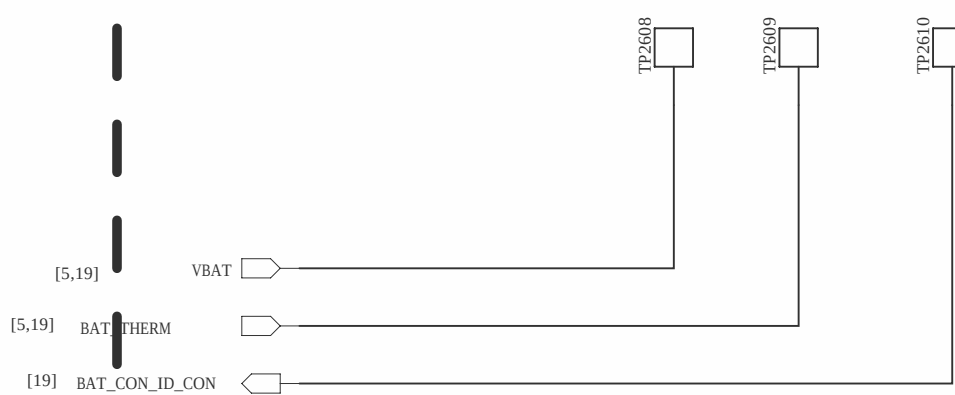
	ID0	ID1	ID2	ID3
	R2901	R2902	R2903	R2904
Bach_AL00	NC	10K	10K	NC
Bach_W09	NC	10K	10K	NC
Bach_L01	NC	10K	10K	NC
Bach_L03	NC	10K	10K	NC
Bach_L09	NC	10K	10K	NC

USB Test Point

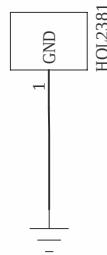


Add a sperate GND close to USBUSB interface if space allowable

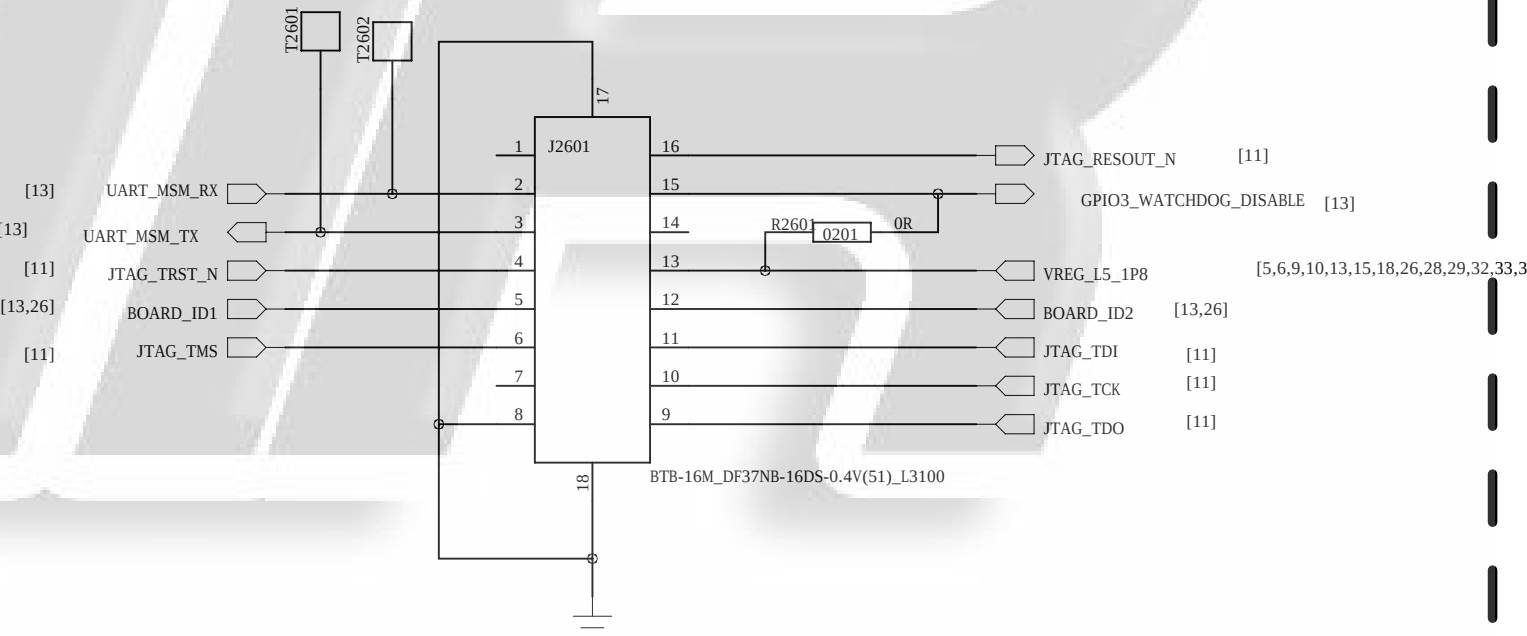
VBAT Test Point



Add BAT_CON_ID of TP if battery has ID pin



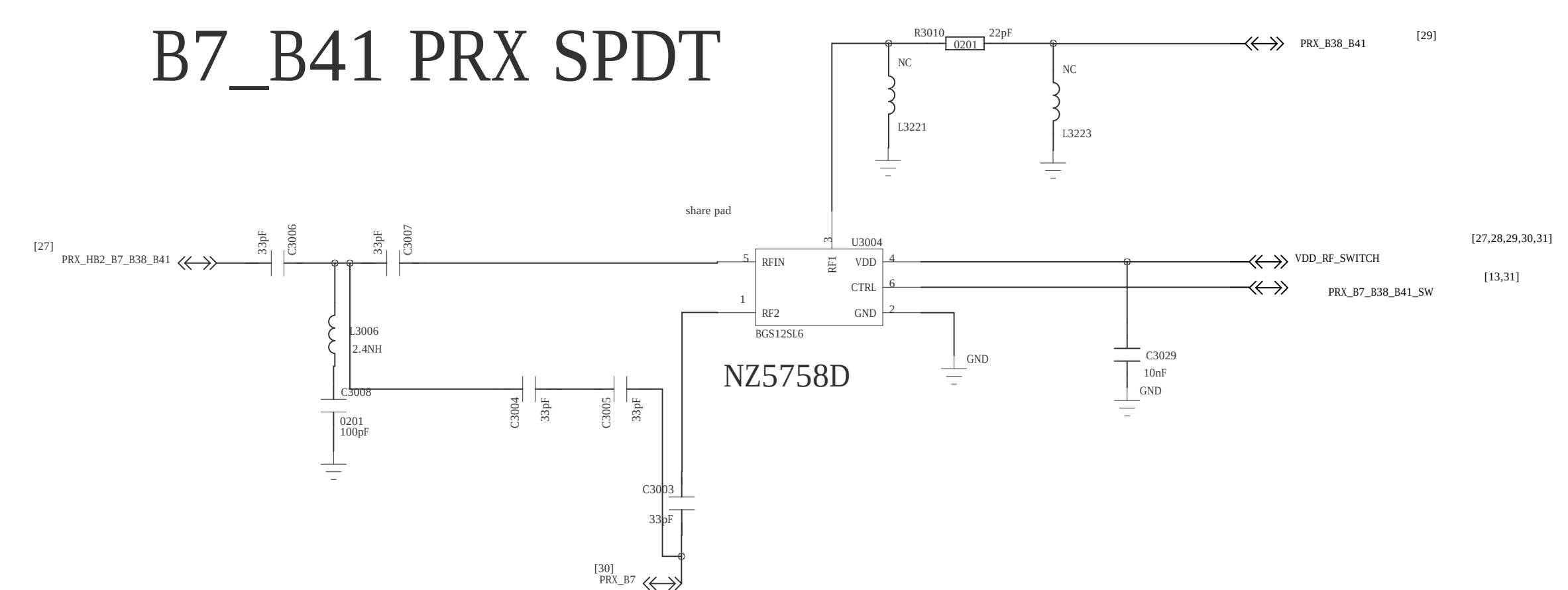
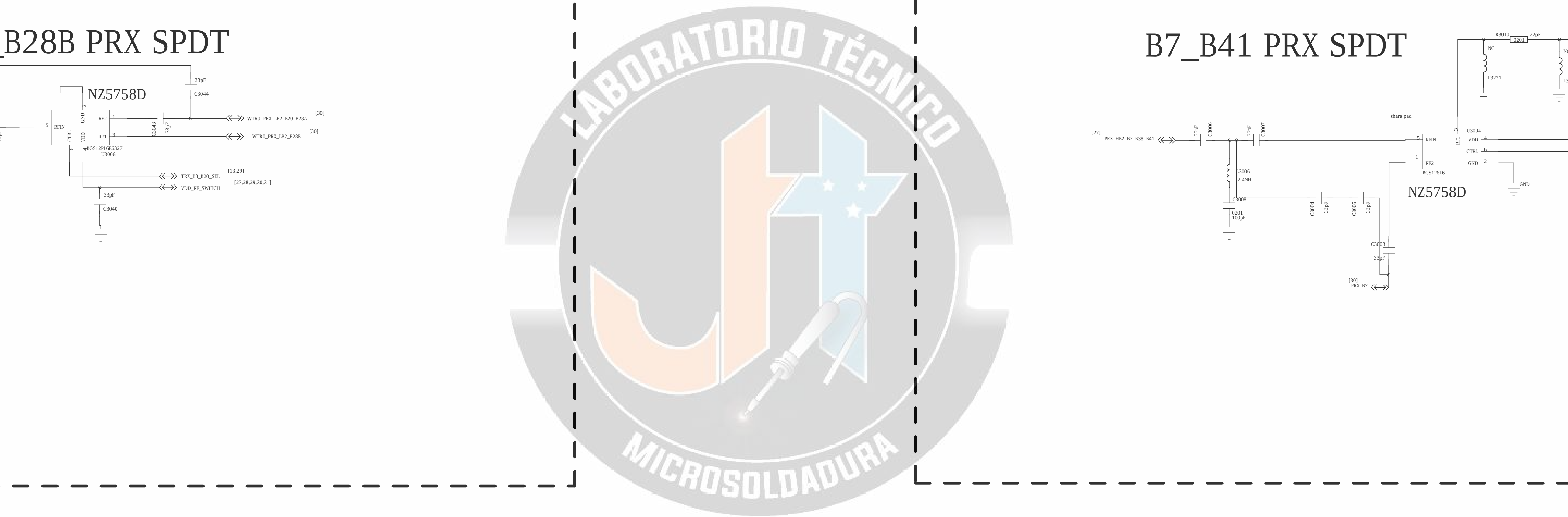
JTAG/UART/BOARD_ID

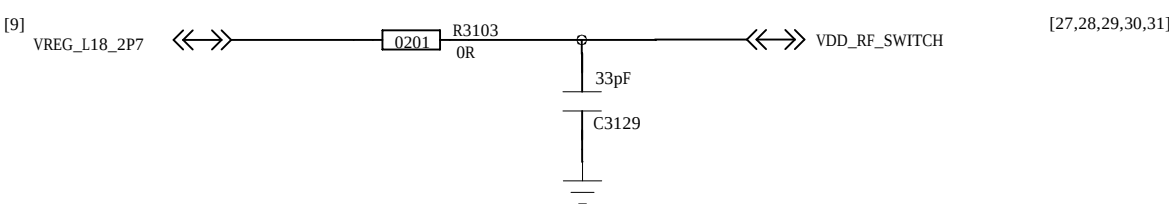


ID0-ID1 \ ID2-ID3	PD-PD	PD-NP	PD-PU	NP-PD	NP-NP	NP-PU	PU-PD	PU-NP	PU-PU
PD-PD									
PD-NP									
PD-PU						Bach-AL00	Bach-W09	Bach-L01	Bach-L03
NP-PD						Chopin-AL00	Chopin-W09	Chopin-L01	Chopin-L03
NP-NP									
NP-PU									
PU-PD						TRT-AL00	TRT-TL10	TRT-L21	TRT-L22
PU-NP									TRT-L23

COMPANY: Shanghai Longcheer Technology Co., Ltd			
TITLE: TEST_POINT			
CODE: XX	SIZE: A0	DRAWING NO: XXXX	REV: A
SCALE: <Scale>			
SHEET: OF 26 OF 36			

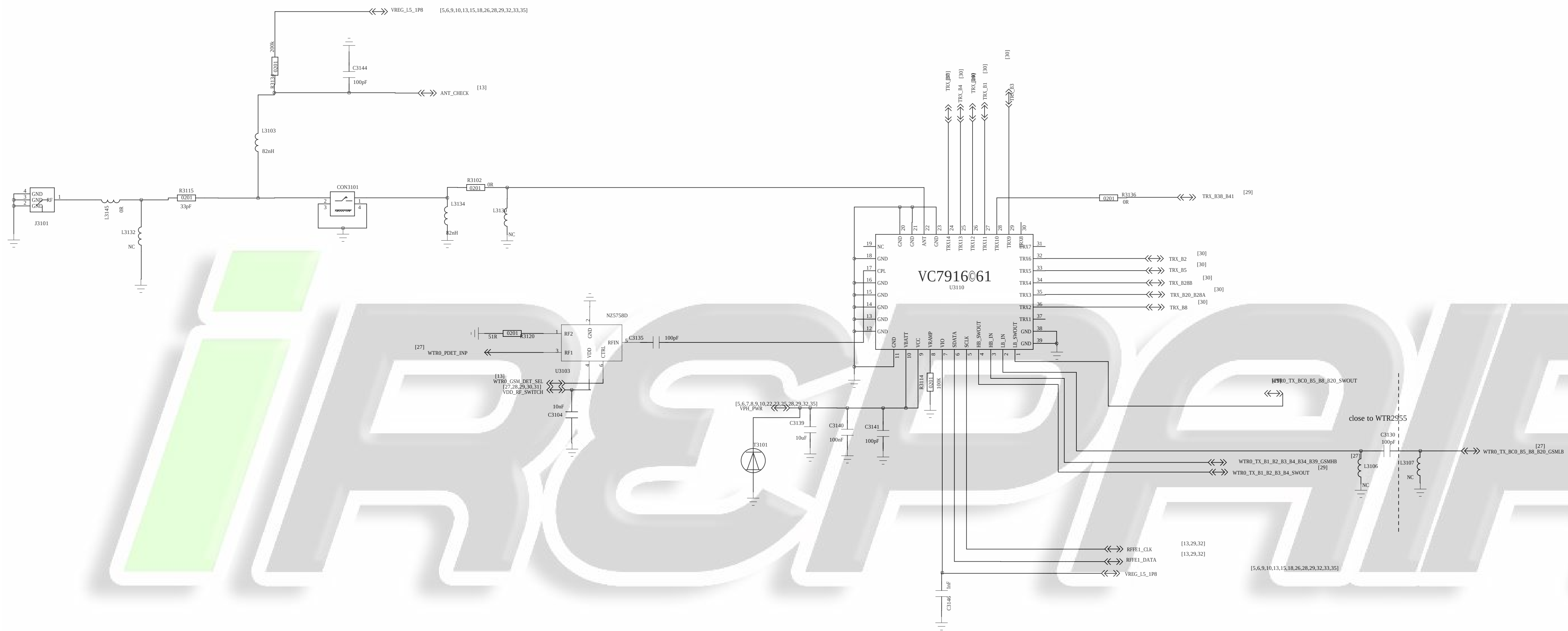
DESIGNER: zhangzhiqiang	DATE: 20160926
CHECKER: yaogang	DATE: 20160926



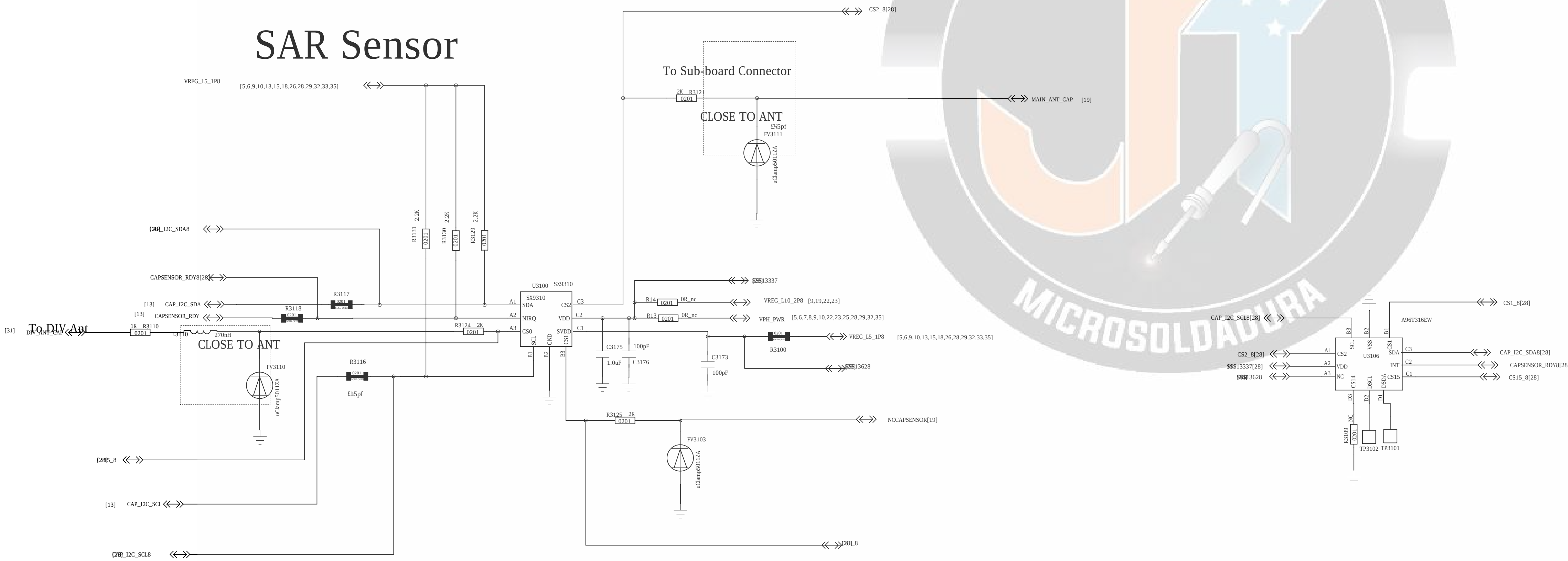


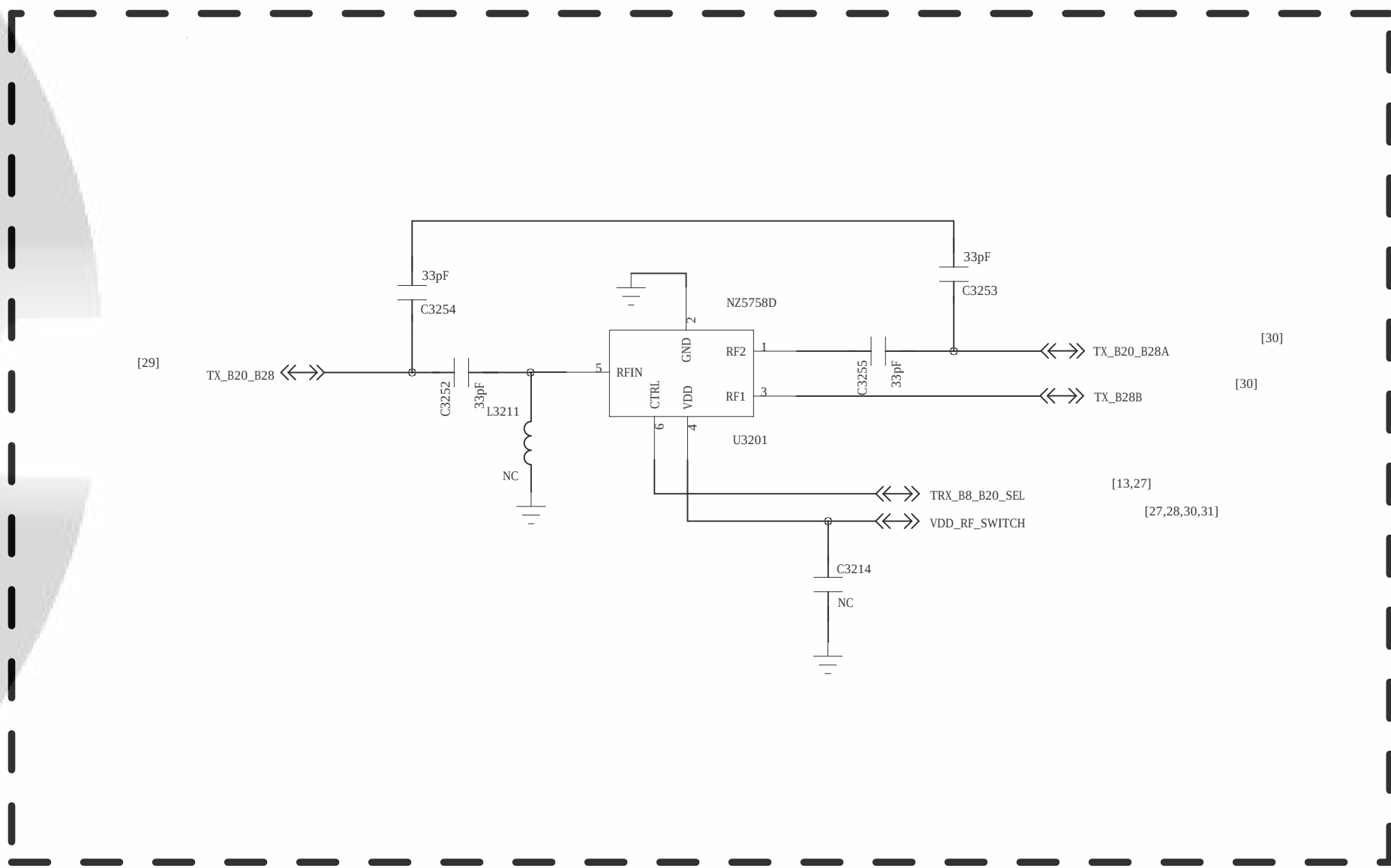
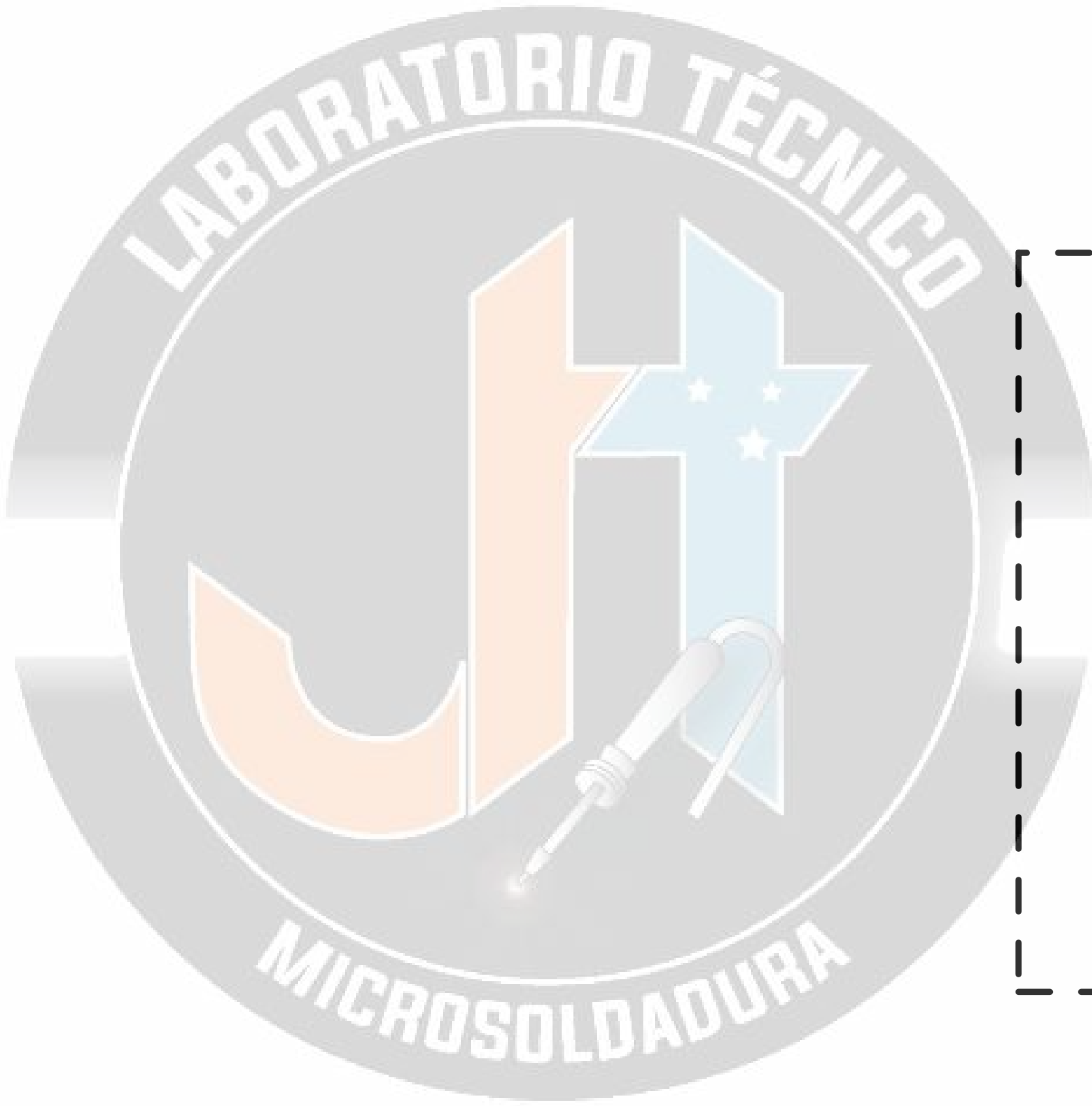
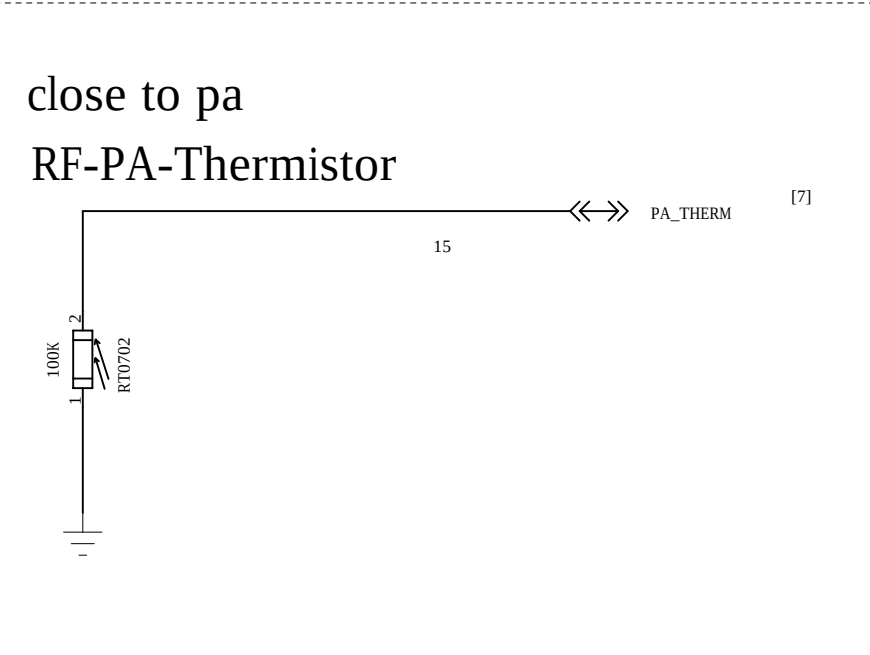
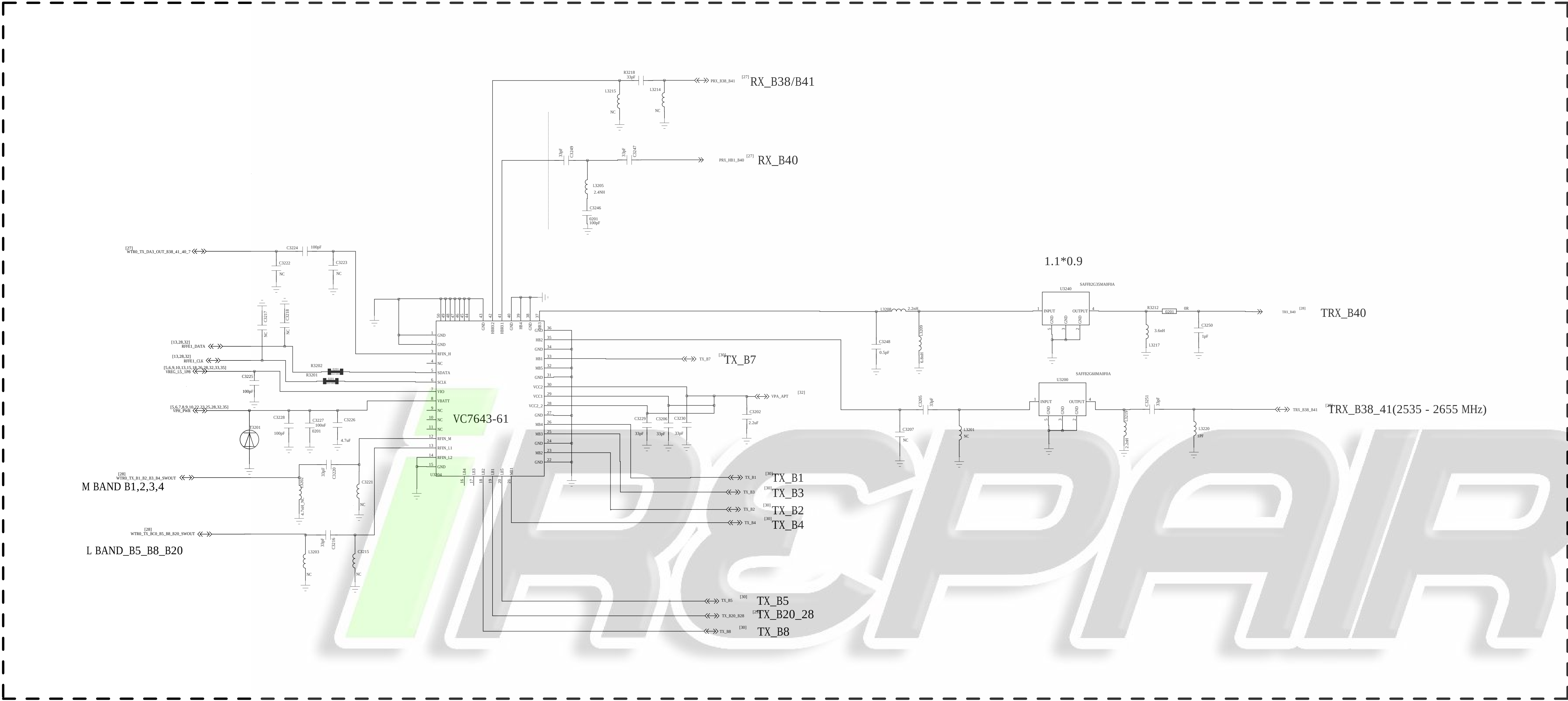
Antenna cut forward

main antenna



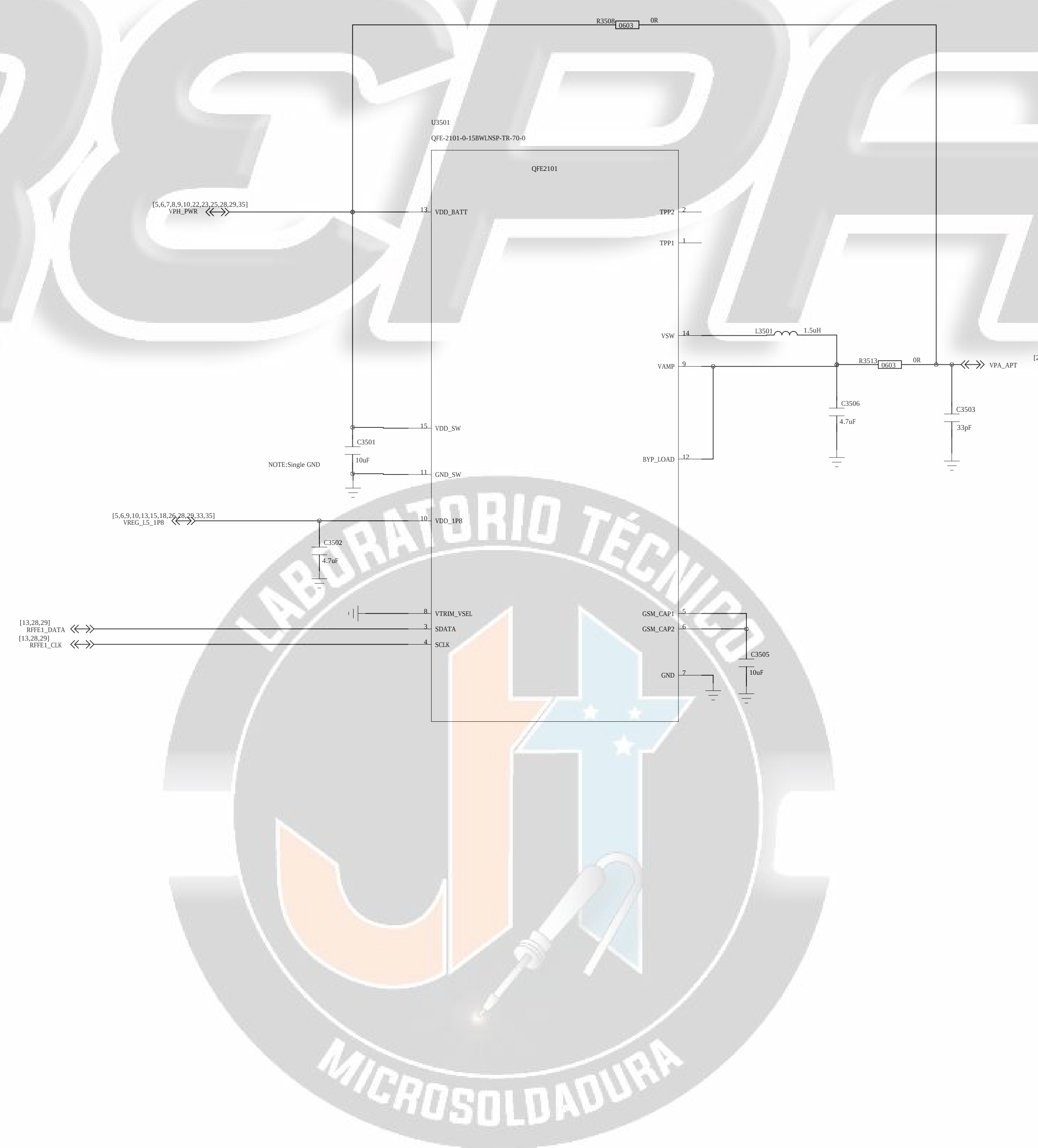
SAR Sensor

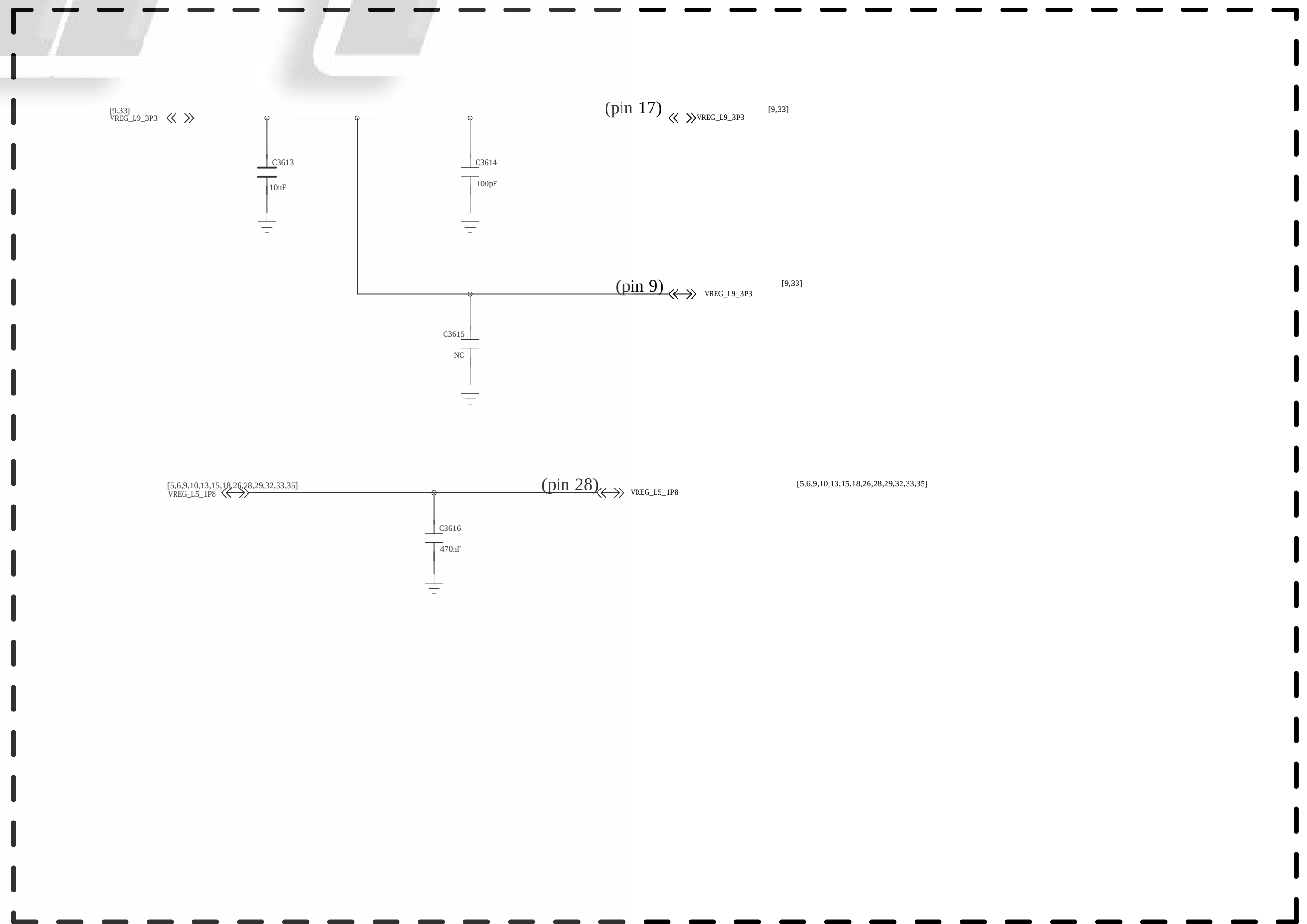
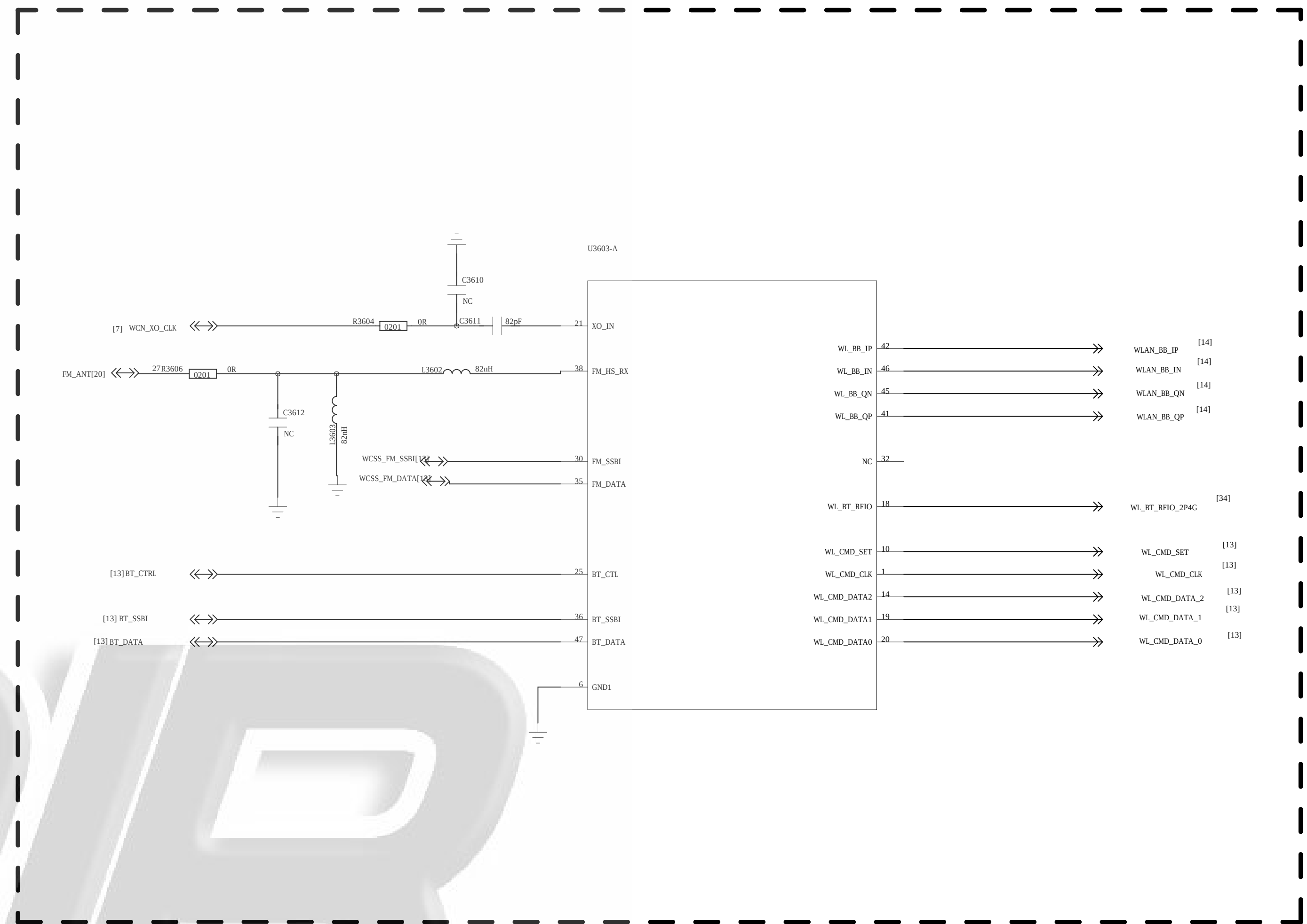
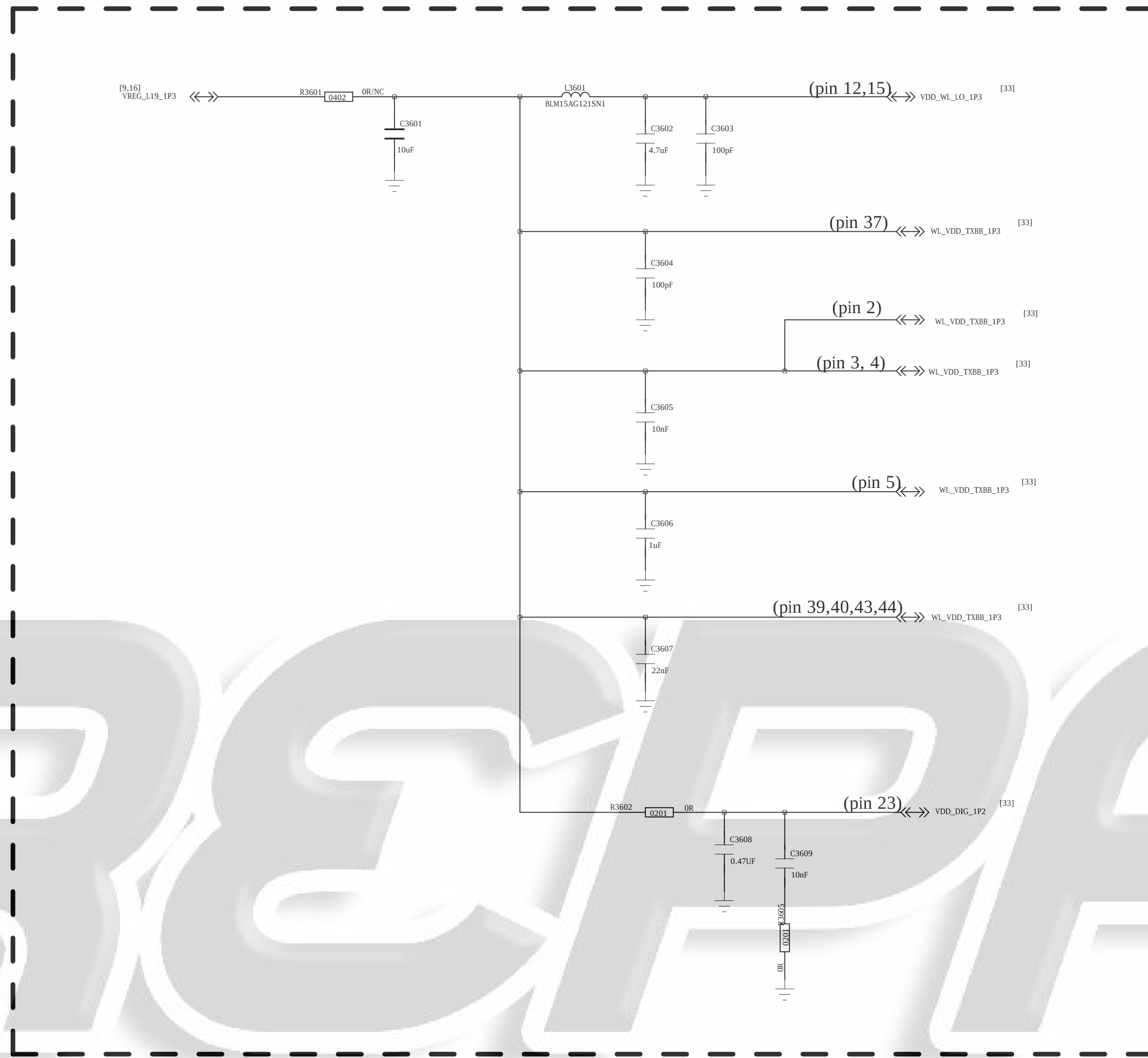
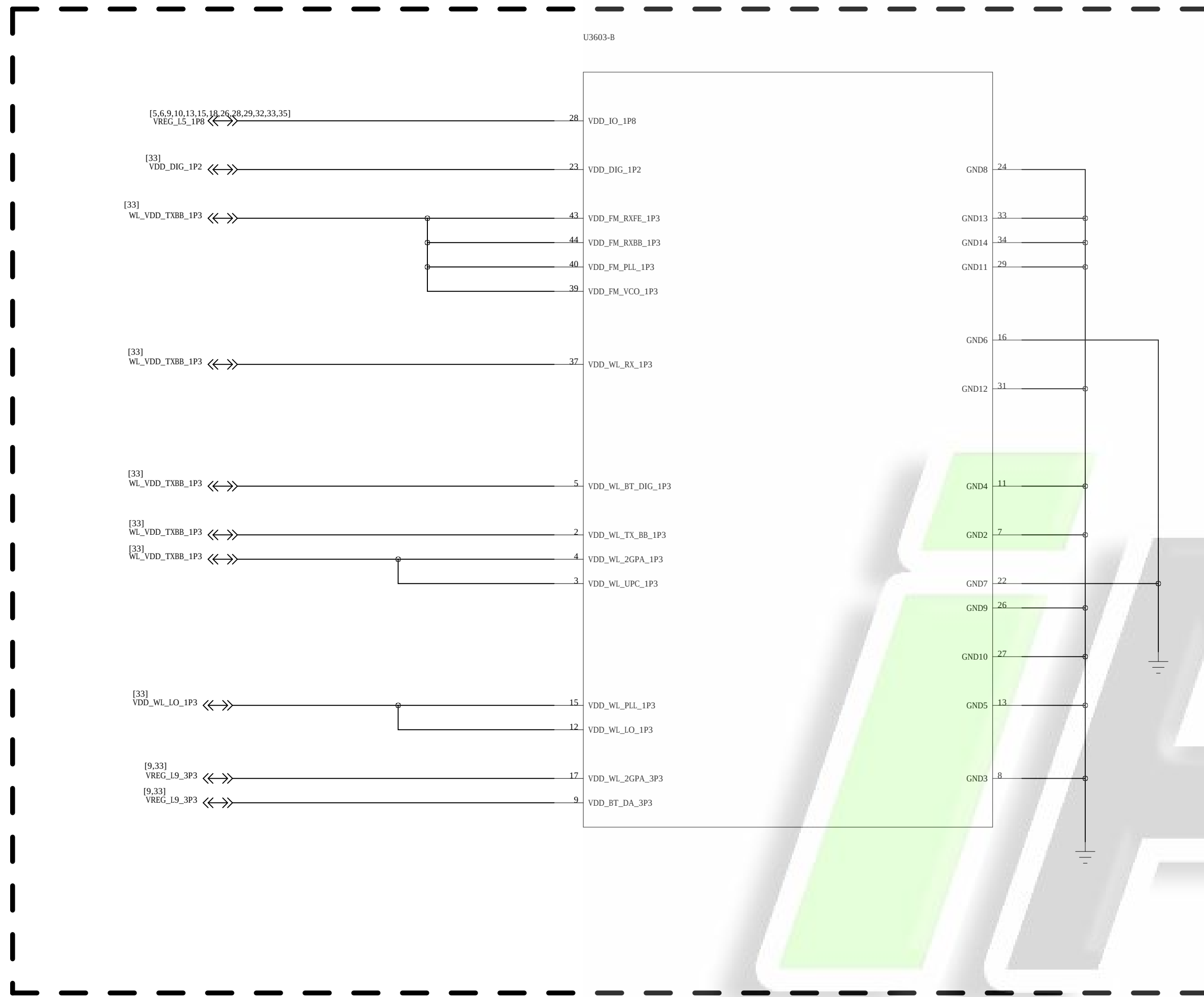




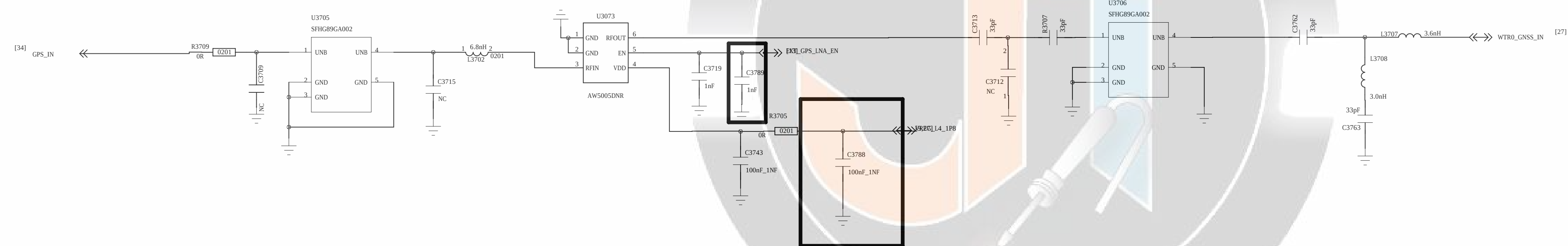
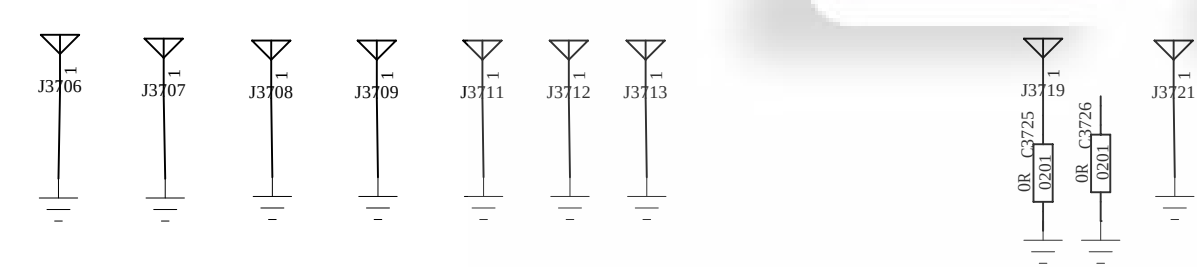
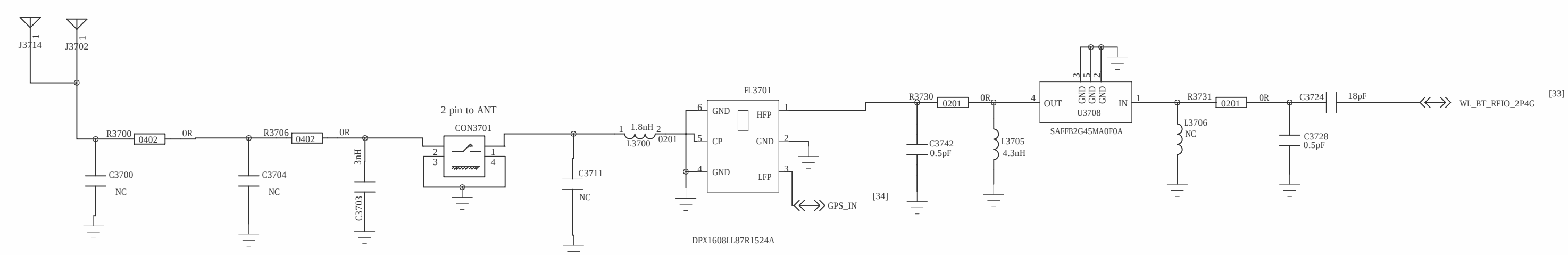
iREPAIR

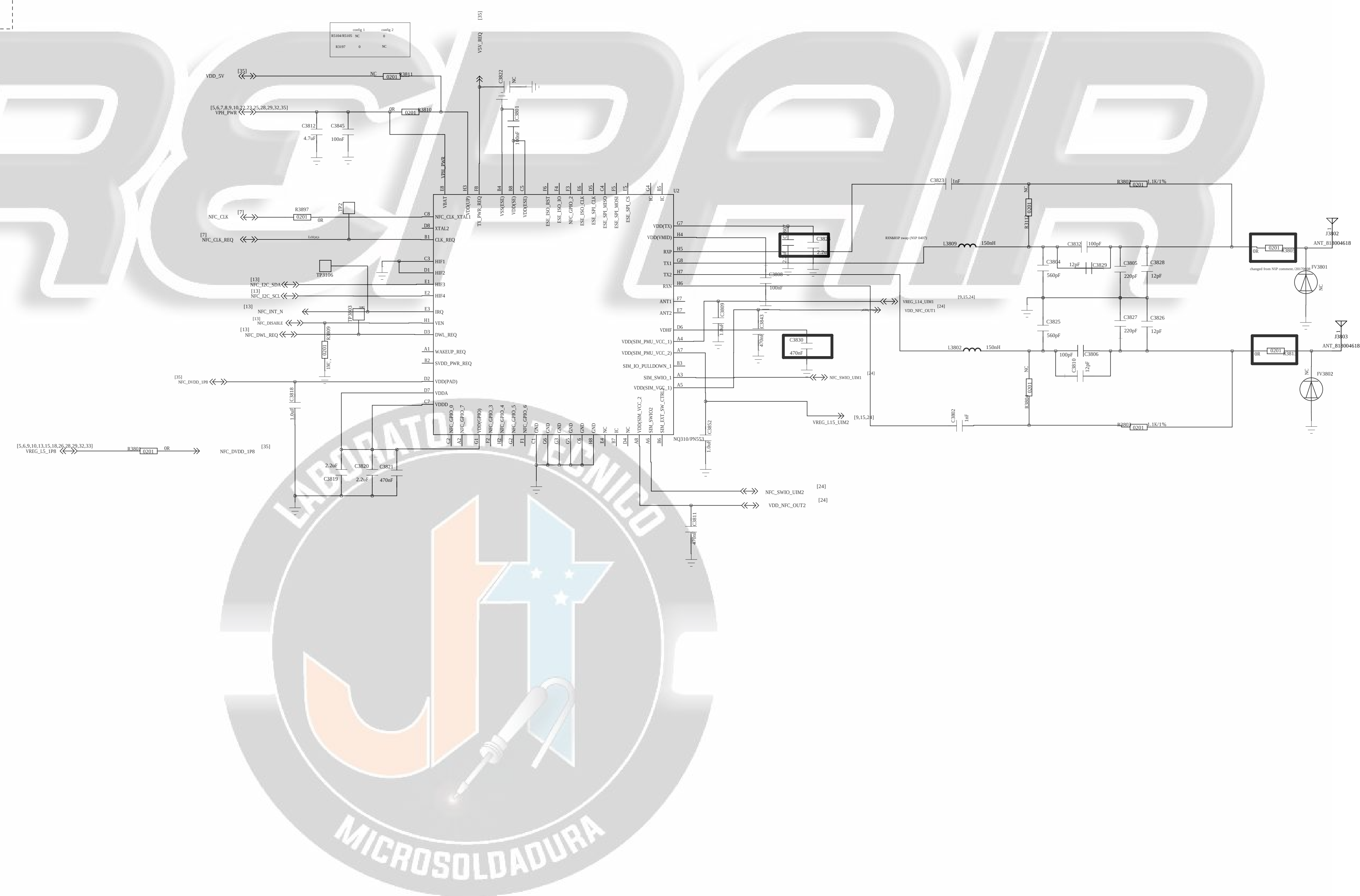
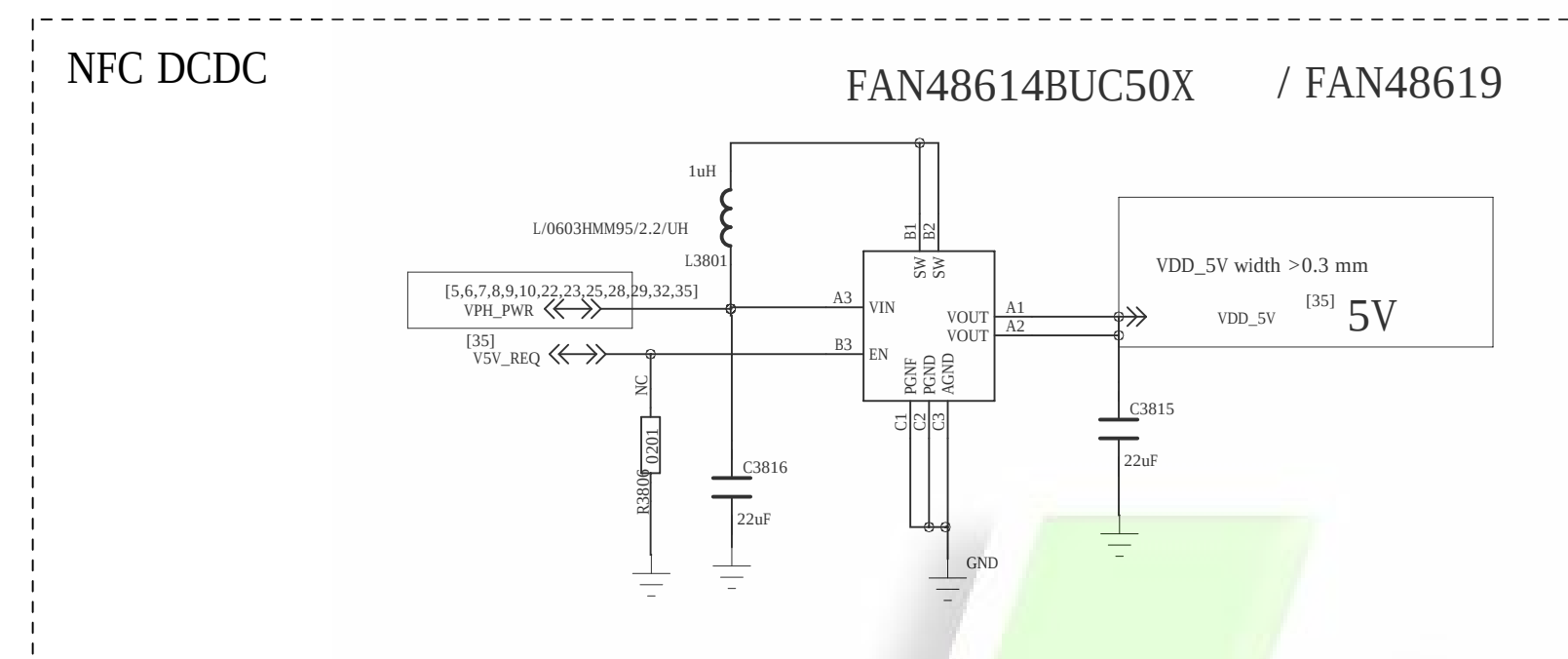
APT





GPS WIFI 2.4G Ant





REVISION RECORD			
LTR	ECO NO.	APPROVED	DATE

iREPAIR



		COMPANY: Shanghai Longcheer Technology Co., Ltd				
		TITLE: 37_DIV_ANT				
DRAWN: zhujinchun	DATE: 20160926					
CHECKED: Lili	DATE: 20160926					
QUALITY CONTROL: <QC By>	DATE: <QC Date>	XX	A0	XXXX	A	
RELEASED: <Released By>	DATE: <Release Date>				SHEET: 06	36